

Appendix B – Case study of lifting operation categorisation
Example activity – Lifting of a generating set and fuel tank with a lorry loader

Situation 1 Straightforward off-load on to level ground beside lorry.			
Environment – 1	Load – 1	Example hazards	Example control measures
Lift categorisation – Basic 		Overturning of lorry loader or failure through overloading of the lorry loader attachment or lifting accessory. Overturning of the lorry loader through ground bearing failure.	■ Accurately assess the weight of the load. ■ Accurately assess the maximum radius at which the load can be lifted. ■ Correctly select the lorry loader and lifting accessories. Assessment of ground conditions and use of spreader pads.
Lifting team ■ Competent person (lift planner) prepares generic risk assessment and method statement. ■ Operator takes the role of lift (crane) supervisor, slinger/signaller and operator.		Planning requirements ■ Generic risk assessment and method statement. ■ On-site review of risk assessment and method statement by lift (crane) supervisor.	

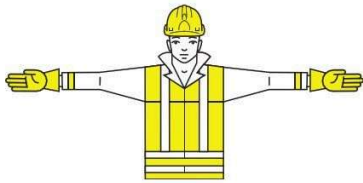
Situation 2 Lifting a generating set and fuel tank in a pedestrian area with narrow access and restricted view of load.			
Environment – 2	Load – 1	Example hazards	Example control measures
Lift categorisation – Intermediate 		Overturning of lorry loader or failure through overloading of the lorry loader attachment or lifting accessory. Overturning of the lorry loader through ground bearing failure. Operator has restricted view of load. Ingress of personnel. Lifting near or over persons.	■ Accurately assess the weight of the load. ■ Accurately assess the maximum radius at which the load can be lifted. ■ Correctly select the lorry loader and lifting accessories. Assessment of ground conditions and use of spreader pads. Use separate slinger/signaller (including use of two-way radio if required). ■ Cordon area off with physical barriers. ■ Utilise site personnel to monitor the area. ■ Close area to public access. ■ Perform lift out of hours.
Lifting team ■ Competent person (lift planner) prepares task and/or site-specific risk assessment and method statement. ■ Operator takes the role of lift (crane) supervisor and operator. ■ Separate slinger/signaller.		Planning requirements ■ Task and/or site-specific risk assessment and method statement. ■ On-site review of risk assessment and method statement by lift (crane) supervisor. ■ Cordoning off of lift area by site manager. ■ Agreement between operator and site manager to fulfil duties of controlling cordoned off area.	

LIFTING OPERATIONS - APPENDIX B

Situation 3 Lifting a generating set and fuel tank in an electrical sub-station with overhead cables and unpaved ground.			
Environment - 3	Load - 1	Example hazards	Example control measures
Lift categorisation – Complex 		Overturning of lorry loader or failure through overloading of the lorry loader attachment or lifting accessory.	■ Accurately assess the weight of the load. ■ Accurately assess the maximum radius at which the load can be lifted. ■ Correctly select the lorry loader and lifting accessories.
		Overturning of the lorry loader through ground bearing failure.	Assessment of ground conditions and use of spreader pads.
		Ingress of personnel. Lifting near or over persons.	Cordon area off with physical barriers. Utilise site personnel to monitor the area.
		Electricity cables.	Have power switched off.
		Blind lift.	Dedicated slinger/signaller.
Lifting team ■ Competent person (lift planner) prepares site-specific risk assessment and method statement. ■ Separate lift (crane) supervisor oversees lifting operation. ■ Dedicated slinger/signaller required. ■ Operator takes the role of operator only.		Planning requirements ■ Site-specific risk assessment and method statement. ■ On-site review of risk assessment and method statement by lift (crane) supervisor. ■ Cordoning off of lift area by site manager. ■ Agreement between operator and site manager to fulfil duties of controlling cordoned off area.	

Appendix C - Recognised hand signals

Start



Stop



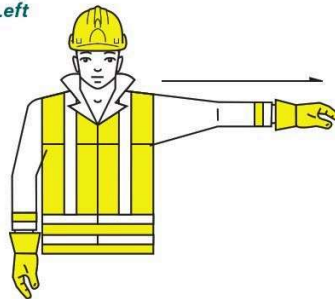
Raise



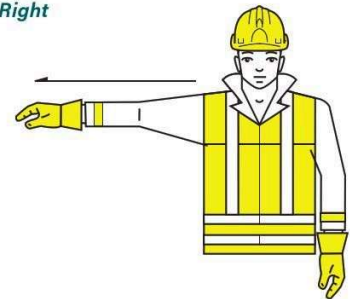
Lower



Left



Right



Horizontal distance



Vertical distance



Move backwards



Move forwards



Danger



End



The signaller should stand in a secure position, where they can see the load and can be seen clearly by the lifting equipment operator, and they should face the operator if possible. Each signal should be distinct and clear. These signals have been reproduced from BS 7121-1 *Safe use of cranes - General*.



For further information refer to the HSE publication *Safety signs and signals. The Health and Safety (Safety Signs and Signals) Regulations 1996. Guidance on regulations (L64)*.

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Appendix D - Wind strengths and effects (Beaufort scale)

Wind force number	Description of wind	Wind effect locally	Speed (mph)	Speed (m/sec)
0	Calm	Calm, smoke rises vertically.	1	0-1
1	Light air	Direction of wind shown by smoke drift, but not by wind or weather vanes.	1-3	1-2
2	Light breeze	Wind felt on face. Leaves rustle. Wind or weather vanes move.	4-7	2-3
3	Gentle breeze	Leaves and small twigs in constant motion. Wind extends light flags.	8-12	3-5
4	Moderate breeze	Wind raises dust and loose paper. Small branches move.	13-18	5-8
5	Fresh breeze	Small trees in leaf begin to sway. Little crested wavelets form on inland waters.	19-24	8-11
6	Strong breeze	Large branches in motion. Umbrellas used with some difficulty.	25-31	11-14
7	Near gale	Whole trees in motion. Becoming difficult to walk against the wind.	32-38	14-17
8	Gale	Twigs break off trees. Progress is generally impeded.	39-46	17-21
9	Strong gale	Chimney pots, slates and tiles may be blown off. Other slight structural damage may be caused.	47-54	21-24

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Lifting equipment



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

There are a number of serious accidents in the construction industry each year as a result of the failure of lifting equipment or accessories.

The majority of these accidents could be avoided through appropriate levels of planning, suitable selection, maintenance, inspection and thorough examination of equipment and comprehensive training, assessment, supervision and briefing of personnel.

This chapter outlines some of the different types of lifting equipment and lifting accessories and gives an overview of the requirements for installing, operating, checking, maintaining, inspecting and thoroughly examining them.

8.1 Introduction

Lifting equipment is any work equipment (such as mobile or static cranes, hoists, telehandlers and excavators) used for lifting or lowering loads (including people) and includes the attachments used for anchoring, fixing or supporting them, such as tie bars used for securing cranes to the cores of high-rise buildings.

Lifting equipment includes any lifting accessories that attach the load to the equipment, in addition to the equipment that carries out the actual lifting function.

Experience has shown that the use of lifting equipment can be a hazardous activity if not properly planned and carried out. The safe use of lifting equipment and accessories will depend upon the following.

- Thorough pre-planning of each lifting operation and the selection of suitable equipment and accessories.
- The availability of suitable lifting equipment and accessories, which are properly inspected and maintained.
- The provision of adequate information, instruction and training on the use of lifting equipment for everyone involved.
- The thorough examination of lifting equipment and accessories, in line with established safe systems of work.

This chapter is intended to provide advice to non-specialists on the implications that utilising lifting equipment and accessories on site will have for them. The content is therefore restricted to information considered relevant for site managers, project managers and others who may become involved in hiring lifting equipment or managing its safe use on site.

As a consequence, the inclusion of technical information on, for example, the technicalities of lifting equipment, its construction, maintenance, inspection and thorough examination, has been limited.

By way of illustration, whilst it is reasonable to expect a site manager to know when a piece of lifting equipment should be thoroughly examined, detail of the content of such examinations is beyond the scope of this chapter.



Problems with lifting operations often occur because of poor equipment and accessory selection.



The general principles of planning and carrying out lifting operations are covered in Chapter C07 Lifting operations.

8.2 Important points

- The rated capacity (safe working load (SWL)) of any item of lifting equipment or lifting accessory must never be exceeded.
- All lifting equipment and lifting accessories must be adequately maintained.
- All lifting equipment and lifting accessories must be subjected to a schedule of inspections and thorough examination.
- Detailed attention must be given to site features (such as overhead cables, unstable ground conditions and adjacent properties).



The collapse, overturning or failure of any load-bearing part of a hoist, crane or other lifting equipment is a reportable dangerous occurrence, even if nobody is injured.

For further information refer to Chapter A13 Accident reporting and investigation.

8.3 Legislative requirements

A brief summary of the main legislative requirements for lifting equipment is listed below.



For detailed information on legislation refer to Section A: Legal and management and for other related legislation refer to Chapter C05 Work equipment and hand-held tools.

8.3.1 Provision and Use of Work Equipment Regulations 1998 (PUWER)

PUWER are concerned with such matters as the selection of suitable work equipment, safeguarding of dangerous parts of machinery, provision of appropriate controls, training of operators, inspection and maintenance of work equipment, including lifting equipment and accessories. PUWER place duties on any person who has control to any extent of any of the following.

- Work equipment.
- A person who uses, manages use or supervises work equipment.
- The way in which work equipment is used at work (including maintenance).

PUWER apply to employers in respect of work equipment provided for, or used by, their employees, self-employed persons in respect of work equipment they use and other people (such as visitors).



For further guidance on PUWER refer to the HSE's ACoP and guidance *Safe use of work equipment* (L22).

8.3.2 Lifting Operations and Lifting Equipment Regulations 1998 (LOLER)

LOLER deal with the specific risks arising from the use of lifting equipment. LOLER build upon PUWER in many areas and apply to all lifting operations, equipment and accessories. LOLER requirements are shown below.

- Specific requirements are applied to the lifting of people.
- Pre-use checks and inspections at appropriate intervals are carried out on lifting equipment.
- All operations are planned, supervised and carried out safely.
- Lifting equipment must be thoroughly examined by a competent (appointed) person on specified occasions and at specified intervals.



For further guidance on LOLER refer to the Health and Safety Executive's (HSE's) ACoP and guidance *Safe use of lifting equipment* (L113).

8.4 Mobile cranes (wheeled and crawler mounted)

A mobile crane is a form of lifting equipment, mounted on a wheeled or crawler chassis, that can move under its own power between lifting positions.

- Wheeled mobile cranes are generally mounted on either truck crane chassis or all-terrain chassis. Truck cranes are generally restricted to travelling on metalled roads and well-compacted ground on site whilst all-terrain cranes can travel on public roads and over most site conditions. Some all-terrain wheeled cranes may have a limited ability to lift and travel with a load (pick and carry duties). Wheeled mobile cranes range in capacity between 25 tonnes and 1,200 tonnes.
- Crawler mobile cranes require transporting to site in component form or on a low loader. Once the crane has been rigged, it can travel around the site with ease and lift and travel with a load. Crawler mobile cranes range in capacity from 40 tonnes to 3,000 tonnes.
- Mini or compact cranes are generally mounted on small crawler or wheeled chassis and may be fitted with outriggers to provide stability. Some are fitted with cabs for the operator and have pick and carry duties. Mini crane capacity is generally in the range of one tonne to ten tonnes.



Crawler crane

LIFTING EQUIPMENT

8.4.1 Planning and use of mobile cranes

The planning and carrying out of lifting operations with mobile cranes should be carried out in accordance with Chapter C07.



For further detailed guidance refer to the *Code of Practice for safe use of cranes – Part 3: Mobile cranes (BS 7121-3)*.

8.4.2 Maintenance, checks and inspections of mobile cranes



For further information on maintenance, checks and inspections refer to Appendix A.

When planning and carrying out maintenance work on mobile cranes, it should not be forgotten that when travelling on the road, the condition of the crane chassis (brakes, suspension, steering, lights, wheels, tyres, and so on) will have a significant effect on the safety of the crane driver, other road users and the public. The Road Vehicles (Construction and Use) Regulations 1986 (Reg. 100) require that they should:



at all times be in such condition ... that no danger is caused or is likely to be caused to any person in or on the vehicle or on a road.

Because a mobile crane has the potential to cause as much damage in a road traffic accident as any other large vehicle, it is important to follow the recommendations set out in the manufacturer's service manual. These normally set maintenance intervals based on engine hours run and distance travelled, whereby the first parameter reached determines the need for maintenance. Failure to maintain the chassis of a mobile crane adequately may put the crane operator and other road users at risk from issues such as those listed below.

- Poor brake performance or failure leading to increased braking distances and an inability to stop in time.
- Insufficient tyre tread, reducing grip, leading to increased braking distances.
- Hydraulic fluid leaks from the suspension system creating hazardous road surfaces and potential loss of control for other road users.
- Tyre blow-outs leading to loss of control of the crane.
- Steering system failures leading to loss of control of the crane.
- Lighting failures leading to the operator not being able to see clearly and other road users not being able to see the crane.
- Wiper, washer and demister failures preventing the crane operator from seeing clearly.
- Failure of the vehicle suspension system.
- Failure of the crane chassis.
- Security of outrigger plates, covers, guards and other potentially loose items.

It is essential that the chassis of a mobile crane is effectively cleaned before maintenance and inspection is carried out, to ensure that defects can be detected.

For further detailed guidance on the maintenance, inspection and thorough examination of mobile cranes refer to the following.



- **Construction Plant-hire Association's *Maintenance, inspection and thorough examination of mobile cranes* (Best practice guide).**
- **Construction Plant-hire Association's *Annual roadworthiness inspection scheme for mobile cranes mounted on a non-standard chassis* (Good practice guide).**
- ***Code of Practice for the safe use of cranes – Inspection, maintenance and thorough examination – Mobile cranes.***

8.4.3 Thorough examination of mobile cranes

The Approved Code of Practice (ACoP) to LOLER states that any testing is required every six or 12 months or at the discretion of the competent (appointed) person carrying out a thorough examination and that the competent (appointed) person will decide on the nature of the test and the method of carrying it out.

Advice from the Health and Safety Executive (HSE) confirms that four yearly overload testing is no longer a legal requirement and repeated overload testing itself may be dangerous as it may create the conditions for additional faults or even failure during testing.

In the HSE's view, the most appropriate method of assessing the condition of a mobile crane is to adopt a defined scope for thorough examination. This is where a specific examination schedule is drawn up for the machine being examined by a competent (appointed) person, which details the parts of the crane that should be thoroughly examined and the supplementary reports and tests that should be carried out.



For further information on thorough examination refer to Appendix B.

8.5 Tower cranes

Tower cranes fall into three main categories.

Top slew tower cranes, which are controlled from a cab at the top of the mast and are erected and dismantled using a suitably sized mobile crane, for which provision has to be made in terms of space and stable ground conditions. Top slew tower cranes are further subdivided into saddle jib and luffing jib types.

Self-erecting tower cranes, which are controlled from ground level, often using remote controls, and are self-erecting and folding.

Vehicle mounted tower cranes, which are often controlled remotely from the ground and should be treated as a mobile crane.



Two saddle jib cranes and a luffing jib crane

8.5.1 Tower crane installation

When contemplating the erection of a tower crane on site, the temporary works co-ordinator will be responsible, and the following points should be taken into account.

- All tower cranes require suitable foundations. For top slew tower cranes particularly, both static and travelling, the foundations will require careful design by a competent engineer.
- Some tower cranes are tied to an adjacent structure for support. The ties and their connections to the crane and the supporting structure will require careful design by a competent engineer.
- On sites where more than one tower crane is erected, there may be overlapping arcs of operation when slewing; anti-collision devices may be fitted or crash radios utilised. As a result, the cranes should be erected with their jibs at different heights and crane slewing operations will have to be co-ordinated.
- The required airspace must be confirmed with regard to the proximity of adjacent structures, overhead power lines or other obstructions.
- The requirement for airspace may need to be discussed with any local airport to establish that there will be no intrusion into, or unacceptably near to, aircraft flight paths.
- There may be a requirement for an aircraft warning light or beacon on the highest point of the crane; the colour intensity and whether it is on steady or flashes will depend upon the local rules for any flight path affected.
- Electrically-powered tower cranes will require a heavy-duty power supply and the crane must be effectively earthed to protect against electrical faults and lightning strikes, including earthing of the appropriate rails. The requirements of the Electricity at Work Regulations apply to all electrically powered tower cranes and their power supply.

8.5.1.1 Over-sailing

In many cases, the crane jib has the capacity to over-sail adjacent properties and areas to which the public have access. Over-sailing is classed as trespass.

Permissions have to be obtained from the respective landowner and a licensing agreement put in place. When drawing up a licensing agreement the following issues need to be addressed.

- What height will the crane operate at?
- Where will the crane be erected?
- What will the turning circle of the crane be?
- What fees will be payable for the consent?
- What times of day (24 hours) will the crane be in operation?
- Where the right to over-sail adjacent property is not given, it may be necessary to:
 - rearrange siting of the tower crane(s)
 - review the type of crane required (for example, a luffing jib crane that has a smaller radius of operation might be the only solution).
- What will the time frame of the licence be?
- How will the crane be erected and dismantled?
- What compliance is needed with applicable laws and regulations?
- What situations would incur termination of the licence?

www For detailed guidance on the design and installation of tower crane bases and ties refer to the Construction Industry Research and Information Association's *Tower crane foundation and tie design* (C761).

www For further detailed guidance on the erection, climbing and dismantling of tower cranes refer to the *Code of Practice for safe use of cranes - Part 5: Tower cranes* (BS 7121-5) and the Construction Plant-hire Association's *Climbing of tower cranes* (Best practice guide).

LIFTING EQUIPMENT

8.5.2 Tower crane operation

Detailed information for carrying out lifting operations with tower cranes can be found in Chapter C07.

In summary, the following points should be taken into account.

- High winds can temporarily stop tower crane operations; each crane should be fitted with an anemometer (wind-speed indicator).
- In high winds, tower cranes must be left in free slew with their hooks raised.
- A plan for rescue from height must be in place in the event of an accident or incident (for example, the operator becoming ill).
- The emergency services have no obligation to carry out these rescues and are often not able to assist because of the height of the rescue and because of restricted access for large vehicles (such as fire service turntable ladders).
- A system must be in place to control the operator hours, taking into consideration the need for operators to take regular breaks to maintain high levels of concentration. BS 7121 suggests the following number of operators.
 - One tower crane: two operators.
 - Two tower cranes: three operators.
 - Three tower cranes: five operators.



For further detailed guidance refer to the Construction Plant-hire Association's *Good Practice Guide, TCIG 2101: Emergency Action Planning & Rescue From Height on Tower Cranes*.

Tower cranes must be secured against unauthorised access, including climbing the tower and unauthorised use.

A lifting plan, approved by the competent person (lift planner), must be in place.

For further detailed guidance on the safe use of tower cranes refer to:



- **Code of Practice for safe use of cranes – Tower cranes (BS 7121-5)**
- **Construction Plant-hire Association's (Best practice guide):**
 - *Safe use of top slew tower cranes (TCIG 0701)*
 - *Safe use of self-erecting tower cranes (TCIG 0601)*
 - *Tower crane operator's handbook.*

8.5.3 Maintenance, checks, inspections and thorough examination of tower cranes



For further information refer to Appendices A and B.



For further detailed guidance refer to the Construction Plant-hire Association's *Maintenance, inspection and thorough examination of tower cranes (Best practice guide)* and the *Code of Practice for the safe use of cranes – Inspection, maintenance and thorough examination – Tower cranes (BS 7121-2-5)*.

8.6 Lorry loaders

Lorry loaders were originally designed to load and unload goods onto and off the body of the vehicle on which they were mounted so that deliveries and collections did not have to rely on other loading and unloading facilities.

In recent years lorry loaders have developed significantly and are now able to carry out many of the tasks traditionally undertaken by mobile and tower cranes.

When used safely they make a valuable contribution to lifting operations. However, it should not be forgotten that all lifting operations must be planned and executed to the same standard, irrespective of the type of lifting equipment being used.

Unfortunately, over the past few years there have been a significant number of accidents involving lorry loaders, which have included fatalities. These could have been prevented by correct planning, supervision, use and maintenance.

The cost in human suffering resulting from poor planning places a moral obligation on employers to improve safety performance. Even the smallest of accidents will also have a financial cost that will impact on the business.



For further information refer to the Construction Plant-hire Association's *Work at height whilst loading and unloading transport (Best practice guide)*.

8.6.1 Types of lorry loader

A lorry loader is a commercial vehicle or trailer fitted with a loader crane. Loader cranes have a column that slews about its base and a boom system that is attached to the top of the column.

Lorry loaders are rated in terms of load movement and are available in sizes from 0.5 metre tonne to 84 metre tonne.

The term metre tonne represents the capability of the lorry loader. For example, 10 metre tonne is represented by:

- 5 metre radius x 2 tonne safe work limit, or
- 10 metre radius x 1 tonne safe work limit.

When in use, the lorry loader is generally stabilised by two stabilisers mounted on the loader crane base. In the past loader crane controls were mounted and operated at the base of the column, but today they are normally portable, wireless units that give the operator greater flexibility of movement and help to keep the load in view.



Typical lorry loader

8.6.2 Planning and use of lorry loaders

The planning and carrying out of lifting operations with lorry loaders should be in accordance with the principles set out in Chapter C07.

8.6.3 Maintenance, checks, inspections and thorough examinations of lorry loaders



For further information refer to Appendices A and B.



For detailed guidance on the safe use of lorry loaders refer to the Construction Plant-hire Association's *The management of lifting operations with lorry loaders* (Best practice guide).

8.7 Mobile elevating work platforms (MEWPs)

Mobile elevating work platforms (MEWPs) are designed to provide temporary working platforms which, when people are in the work platform, can be easily adjusted for height and outreach.

They can also be easily moved from one location to another, and are particularly suitable for tasks where the use of a ladder would be unsafe and the erection of a scaffolding platform too time-consuming or impracticable in relation to the job to be done.

Good management, planning and consultation with interested parties before and during any work close to overhead lines will reduce the risk of accidents. This applies whatever type of work is being planned or undertaken, even if the work is temporary or of short duration. Stop and consider the risks if the intended area to work in is within a distance of 10 m, measured at ground level horizontally from below the nearest wire.

Remove the risk. The most effective way to prevent contact with overhead lines is by not carrying out work where there is a risk of contact with, or close approach to, the wires. If this cannot be avoided and there is a risk of contact or close approach to the wires, consult the owner to find out if the line can be permanently diverted away from the work area or replaced with underground cables.

This will often be inappropriate for infrequent, short-duration or transitory work. If this cannot be done and there remains a risk of contact or close approach to the wires, find out if the overhead line can be temporarily switched off while the work is being done.

Sizes and capabilities of MEWPs vary considerably. Small, one-person machines are available, with a maximum rated capacity (SWL) of 120 kg, and working heights of a few metres, whilst at the other end of the scale, work platforms may be 4 m x 2 m or more and have a rated capacity (SWL) in excess of 1,000 kg.

Extending boom heights can exceed 100 m and the outreach of some units can approach 40 m.

8.7.1 Types of MEWP

Common types of MEWP found on construction sites are listed below.

- Scissor lifts.
- Telescopic booms.
- Vertical lifts.
- Articulating and multi-boom articulated.

These types of MEWP may be mounted on chassis that are towable units, vehicles, self-propelled (wheeled or crawler) or pedestrian controlled.

LIFTING EQUIPMENT

8.7.1.1 Scissor lift

This type of MEWP gives a substantially vertical lift. The work platform may be fitted with a sliding extension that will give a small amount of outreach.

8.7.1.2 Vertical lift

This type of MEWP gives vertical lift only and is generally used for small, lightweight machines for indoor use only.

8.7.1.3 Telescopic boom

This type of MEWP gives both vertical height and outreach, and the telescoping function gives smooth and fine control near structures. The work platform may also be able to swivel in the horizontal plane and is mounted on a fly jib for additional manoeuvrability.

Boom-type MEWPs are sometimes known as cherry pickers. They generally have smaller baskets and lower lift capacities than scissor-type MEWPs, and their platforms can 'bounce' at height due to the boom structure flexing. This may make them less suitable to use for installing long or heavy materials, or bulky materials that may obstruct the control panel.

8.7.1.4 Articulating and multi-boom articulated

These types of MEWP may be mounted on a road-going vehicle chassis or a self-propelled wheeled or crawler chassis. They give a wide range of reach and height.

Vehicle-mounted MEWPs are nearly always equipped with outriggers and specialised types are available (for example, machines that enable access to the underside of bridge arches from the roadway above).

Units mounted on a self-propelled chassis have a 'travel while elevated' ability and may have four-wheel drive. Rough-terrain MEWPs have been specially developed for construction site work.



Scissor lift



Vertical lift



Telescopic boom



Articulating MEWP

08 8.7.2 Planning and use of MEWPs

The planning and carrying out of lifting operations with MEWPs should be in accordance with the principles set out in Chapter C07. In addition the points below should be taken into account.

8.7.2.1 Entrapment of people on the platform

In certain circumstances (such as working at height amongst steelwork), there is an increased risk to people in the MEWP of becoming trapped and, in some cases, crushed between the platform and a fixed obstruction. There have been a number of such incidents in the UK over the past few years, several of which have resulted in fatalities. Job planning should consider this hazard and, where required, additional measures should be taken.

- Consideration of additional safety devices on the MEWP.
- Devising and implementing a rescue plan for people trapped at height.
- Briefing of operators and other people who will be working from the platform.

This issue has been highlighted by the HSE as a result of several accidents in which MEWP operators had become injured and immobilised and unable to release themselves, having become trapped between the machine (such as the top guard-rail or controls) and a fixed structure (often structural steelwork).



For comprehensive guidance on avoiding entrapment refer to the Strategic Forum for Construction Plant Safety Group's *Avoiding trapping/crushing injuries to people in the platform: Best practice guidance for MEWPs* and the HSE's guidance *Avoiding trapping/crushing injuries to people in the platform of mobile elevating work platforms (MEWPs)*.

8.7.2.2 Emergency controls

Most MEWPs can be controlled either at ground level or from within the platform.

Both of these controls and the emergency stop and emergency lowering controls should be checked as part of the pre-use inspection to ensure they are working correctly.

The operator, or their supervisor, should ensure that a responsible person, who is familiar with the emergency lowering system, is always in close proximity to the MEWP, to lower the work platform in the event that rescue is required.



Auxiliary controls decal

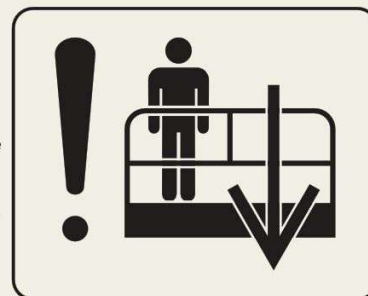
This image can be used on MEWPs, in conjunction with existing manufacturer's symbols, to aid location of the emergency controls.

Even experienced operators sometimes have difficulty locating the auxiliary controls that every MEWP is fitted with.

This decal (sticker) is a practical visual aid and a prime example of an industry initiative to make access equipment even safer.

The decal should be positioned to clearly indicate the location of the auxiliary controls. Employers should ensure that somebody at ground level is properly trained on how to use the controls and can gain access to the controls in an emergency.

Personnel must never attempt to climb out of an elevated MEWP if the emergency back-up system fails to work, but should stay in the work platform until rescued by other means.



Decal for auxiliary controls for descent in an emergency



This decal is available to download free from the IPAF website.

8.7.3 Overturning

- Check for soft ground, drains and other unsuitable ground conditions before deploying the outriggers or stabilisers; check for any hazards before siting the machine. These items should all be considered at the planning stage.
- Beware of overloading, especially if the work platform is being used at maximum outreach to remove fans, motors or other heavy loads. Always observe the rated capacity (SWL).
- Be careful when operating on a slope, even if the machine is properly levelled with the stabilisers or outriggers.
- Always check that the machine is stable before operating.
- Be careful when travelling with the work platform raised, especially on poor ground conditions. Check that the MEWP is designed to be moved in the elevated position.
- Never attempt to travel while the outriggers or stabilisers are deployed, unless the machine is designed for this purpose.
- Be careful not to collide with any obstruction or other vehicle.
- Some modern machines are fitted with a tilt sensor, which, if activated, enables the work platform to only be lowered.



Mobile boom

8.7.4 Fall protection

The following refers predominantly to the use of boom-type MEWPs although, in some circumstances, parts could also be relevant to the use of scissor or other vertical lifts. Personal fall protection systems come in two categories: work-restraint systems and fall-arrest systems. In most instances for MEWPs, the use of work restraint is preferable over fall arrest due to the impact that the forces resulting from a fall could have on the stability of the MEWP.

8.7.4.1 Work-restraint system

Work-restraint equipment prevents the fall from happening. The main feature of work-restraint equipment is that a shorter lanyard is used, which restricts the operator's limit of travel to the confines of the work platform. Lanyards should be carefully selected, taking into account the features of the machine on which they are to be used, to ensure that the user cannot get into a situation where a fall could occur. The lanyard must always be attached to a designated anchor point in the work platform. A work-restraint lanyard must **not** be used for fall arrest.

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8.7.4.2 Fall-arrest system

Fall-arrest equipment allows a person to fall but arrests the fall before the person hits the surface below. Where a decision has been taken to rely on fall-arrest equipment as a means of preventing injury, it will be necessary for the operator to wear a full body harness and a lanyard equipped with an energy absorber. Vital considerations are shown below.

- The height at which work is being carried out must be such that it allows the energy absorber to deploy to arrest the fall before the wearer hits the surface below. A minimum working height of 6.75 m is recommended if fall arrest is to operate successfully.
- The anchor point on the machine must have been designed to withstand the shock-loading of arresting a fall. Many anchor points fitted to MEWPs are only rated for work restraint. If it is not marked with its rating, contact the manufacturer.
- Users of harnesses and lanyards must have been trained in their use, inspection and care.
- In arresting a fall, a boom-type MEWP will flex, which could eject other occupants and materials out of the basket. This could also overturn the machine if it is operating at maximum outreach.
- A check must be made that the structure has no projecting features that the falling person would strike during the fall.
- Decide how the person who has fallen will be rescued after the fall has been arrested. This may be as simple as another operator gently lowering the boom of a MEWP using the ground level controls until the fallen person is at ground level.

An exception to wearing a harness whilst operating a MEWP is when the machine is working over, or near to, water where there is a risk of drowning. In these scenarios a lifejacket should be worn.



IPAF recommends a full body harness with a short lanyard in boom type platforms

 Visit the IPAF website to download free Clip on! decal stickers.

 For advice on the inspection of harnesses and lanyards refer to the HSE information sheet *Inspecting fall arrest equipment made from webbing or rope (INDG367)*.

8.7.5 Working over or near to water

There could be a danger of drowning where a MEWP, particularly an articulating boom (cherry picker), is being used over, or near to, water. It is possible that the work platform could come to rest underwater if it were to overturn. It is essential, therefore, that the occupants do **not** clip on to the anchorage point, and do wear appropriate personal flotation equipment (such as a lifejacket). In such circumstances:

- even though the machine toppling would have a traumatic effect on the user(s), they still might have a chance of being able to swim to, or scramble back onto, dry land
- a risk assessment must be carried out to determine and implement effective fall prevention measures that would normally be covered by wearing a harness and lanyard.

 For further information refer to Chapter F05 Working over or near to water.

8.7.6 Restricted or enclosed spaces

- Extra care is needed when working in, or manoeuvring into, restricted or confined spaces to avoid collisions and/or entrapment. Operators must have a full understanding of all controls.
- If the MEWP has an internal combustion engine, remember that the exhaust fumes will cause a hazard in any confined or enclosed space.
- Batteries for units should not be charged in enclosed spaces. There is a hazard of explosive hydrogen gases being given off.
- Liquefied petroleum gas (LPG) powered vehicles should not be refuelled in a confined space. Any spillage of fuel will quickly and dramatically expand into a large gas cloud. The gas will then accumulate at the lowest point and create an explosive hazard.

8.7.7 Interference with MEWPs in public places

- Whenever MEWPs are used in public places, additional precautions should be taken to ensure segregation between members of the public and the machine. This may be by the use of barriers or posting marshals around the machine.
- Additional interlocks or guards may be necessary to prevent operation of, or tampering with, ground level controls by unauthorised persons.
- Care should be taken against the risk of entrapment as a result of inquisitive people, and especially children, getting too close or underneath MEWPs. Scissor lifts are particularly hazardous.

For further detailed guidance on the planning and safe use of MEWPs refer to the:

- **Code of Practice for the safe use of MEWPs (BS 8460)**
- **HSE information sheet *The selection, management and use of mobile elevating work platforms (GEIS6)***
- **HSE's online construction safety topic *Mobile elevating work platforms (MEWPs)*.**

8.7.8 Maintenance, checks, inspections and thorough examinations of MEWPs



For further information refer to Appendices A, B and D.

8.8 Construction hoists

Hoists of various types are widely used on construction sites. These range from high-capacity passenger/goods hoists, goods-only hoists, transport platforms, and inclined hoists (often referred to as construction hoists, where the load is guided), to scaffold hoists and trestle/beam hoists and the humble gin wheel (where the load is not guided).

As pressure increases to become more efficient and reduce manual handling, hoists are an increasingly vital part of many construction sites. The smooth and rapid movement of persons and materials assists in efficient management of projects. Even where other methods of moving materials are to be used (for example, by tower crane), it is not unusual to find a hoist used solely to assist the scaffold erection process.

The number and type of scaffold hoists and trestle/beam hoists available for hire has increased significantly and their use will require a lifting plan (refer to Chapter C07) to be drawn up by a competent person (lift planner). This plan must consider possible exclusion zones around the hoist in the event of the load dropping, safe slinging methods, the appropriate lifting accessories, and so on. These are a different set of risks to loads being carried in the enclosed platform of a goods hoist.

Risks can arise from a late decision to use a hoist to solve a previously unforeseen problem during construction. For example, if planning to use a hoist tied to a scaffolding system that is already in place but which was not originally designed to take the additional loadings, there is a need for careful planning and discussions with the scaffold designer before simply allowing a hoist to be hired and fixed to the scaffold.

The requirement for a dedicated power supply to the hoist must also be considered (except for manually-operated hoists and any with an independent power supply). Most are electrically powered, so the Electricity at Work Regulations 1989 will apply. Electrical requirements may result in the need for generators, which must be of adequate capacity. This introduces potential noise, fumes, fuel storage and spillage issues.

8.8.1 Types of hoist

A wide range of hoists are used on construction sites. The main types are described below.

8.8.1.1 Passenger/goods hoists

- Have a rack-and-pinion drive mechanism.
- Are tied to an adjacent structure.
- Can be erected to lifting heights of up to 250 m.
- Can travel at up to 100 m per minute.
- Are operated from inside the cage.
- Available in a range of cage sizes, up to 1.5 m wide x 4.6 m long.
- Available with payloads of up to 3,200 kg.

8.8.1.2 Goods only hoists

- Generally have a rack-and-pinion drive mechanism.
- Are tied to an adjacent structure.
- Can be erected to lifting heights of up to 100 m.
- Can travel at up to 40 m per minute.
- Are operated from outside the cage.
- Available in a range of cage sizes, from 1.4 m wide x 0.8 m long, to 1.5 m wide x 6.0 m long.
- Are available with payloads from 200 kg up to 2,500 kg.



Hoist operators carrying out pre-use checks on a passenger/goods hoist



Pre-use checks should only be carried by authorised personnel on goods-only hoists

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8.8.1.3 Transport platforms

Transport platforms are dual-purpose machines that either carry goods only, in which case they are operated from outside the platform, or carry passengers, in which case they are operated from inside the platform and are restricted to a speed of 12 m per minute.



For detailed guidance on transport platforms refer to the Construction Plant-hire Association's *Safe use of transport platforms* (Good practice guide).



Equipment designed to carry goods only should be marked accordingly

8.8.1.4 Inclined hoists

- Generally have a wire-rope drive mechanism.
- Are supported by an adjacent structure.
- Can be erected to lifting heights of up to 40 m.
- Can travel at up to 25 m per minute.
- Are operated from outside the platform.
- Have platforms of about 0.7 m x 0.8 m.
- Are available with payloads up to 250 kg.

8.8.1.5 Scaffold hoists

Scaffold hoists are not hoists in the true sense, in that the load is not guided, as is the case with a goods or passenger/goods hoist. They are in fact small cranes that are generally attached to scaffolding, and have a small swinging jib on which is mounted a wire rope winch. They:

- have a lifting capacity of up to 250 kg.
- have a standard lifting height of up to 50 m.
- can lift at up to 25 m per minute.



Inclined hoists should be supported by the adjacent structure



Scaffold hoists should only be fixed to designated points on a scaffold

8.8.1.6 Trestle/beam hoists

Trestle/beam or gantry hoists consist of a runway beam mounted in a trestle structure, with a wire rope winch mounted on the beam. The beam is cantilevered over the edge of a building roof or floor, enabling materials to be lifted from ground level. They:

- have a lifting capacity of up to 1,000 kg.
- have a standard lifting height of up to 25 m.
- can lift at up to 20 m per minute.

8.8.2 Planning and installation of hoists

The installation and dismantling of hoists requires good planning and co-ordination and must be undertaken by competent individuals in compliance with legal requirements and relevant standards. Special attention must be given to ground conditions to ensure adequate support for the hoist. The forces imposed on the structure by the hoist, its loads and inclement weather, particularly high winds, must also be taken into account. Hoist operators must be adequately trained, competent and authorised and should be specifically responsible for ensuring that the hoist is not overloaded or otherwise misused.

8.8.2.1 Pre-planning (construction hoists)

When it has been decided (ideally at the tender stage) that a construction hoist will be needed, requirements must be quantified as follows.

- Consideration of the locations for siting the hoist.
- The loads to be carried in terms of size and weight and whether goods only or both goods and passengers are to be carried.
- The likely loading on the structure and whether any enabling work or structural changes are required.
- Any constraints in the position where the hoist can be erected and whether other features (such as scaffolds) must be modified to accommodate the hoist.
- The loading on the hoist's foundations.

At an appropriate point, the user of the hoist and the supplier of the hoist should assign a competent (appointed) person who must liaise with each other to ensure that all aspects of the hoist selection, delivery, erection and use are carried out safely. The user's competent (appointed) person may need to rely upon the experience and competence of the supplier's competent (appointed) person in order to fulfil their responsibilities. The detailed requirements of these roles, both of which require specific training and previous relevant experience, are considered to be outside the scope of this chapter.



For further information refer to the *Code of Practice for the safe use of construction hoists* (BS 7212).

A full and detailed risk assessment should be carried out before a hoist is positioned. A method statement can then be developed, which must then be agreed with the user.

8.8.2.2 Site survey

Prior to delivery of the hoist it will be necessary for a competent person, representing the supplier, and the lift planner to visit the site to establish the practicalities of installing it. During the site survey the supplier's competent (appointed) person will determine the appropriate type of hoist required, based upon the criteria and information on intended usage of the hoist provided by the user's competent (appointed) person.

The findings of the survey will determine the optimum position for the hoist, based upon factors such as those listed below.

- Co-location with materials storage areas.
- Avoidance of hazardous features (such as overhead cables).
- The need for safe access at all levels.
- The need for safe access to the cage or platform at ground level.
- The need for foundations and drainage.
- Suitable locations on the building structure for the attachment of ties.

In selecting the most appropriate hoist, the supplier's competent (appointed) person will need to take into account factors such as those listed below.

- The space constraints of the area.
- Access implications relating to the size and weight of the hoist components.
- Ground conditions and suitable locations on the building structure for the attachment of ties.
- The proximity of site features (such as overhead power lines, railway tracks and adjacent public thoroughfares).
- Other work that will take place whilst the hoist is in place and how the site might change during this period.
- Foreseeable extremes in weather conditions.

8.8.2.3 Erecting, altering and dismantling hoists

Hoists that rest on the ground must only be erected on a firm base, adequately supported and secured. All materials supporting the hoist must be strong enough to support the weight of the completed hoist structure and its maximum load, and be free from defects.

The erection of the hoist must be carried out in accordance with the manufacturer's instructions and a method statement that has been discussed with and approved by the user, as the proposed method and timing may impact upon other site work.



Refer to Chapter C07 Lifting operations.

In many ways, erection, alteration and dismantling are potentially the most hazardous work associated with hoists on site. Anyone not directly involved in erecting the hoist should be excluded from the area.

Planning how the hoist will be erected, and deciding who is responsible for what, are vital when implementing a safe system of work. This is particularly important where the hoist will be built in conjunction with the erection of a scaffold. It is essential that the erection and, later, the modification and dismantling of hoists are properly planned, adequately supervised and carried out in a safe manner by competent people.

Where the hoist is attached to and supported by a scaffold, the scaffold must be designed to take account of the imposed loadings. This becomes more important where the use of a larger capacity hoist is being considered. Hoists should only be erected by competent, authorised personnel who hold a relevant card.

8.8.2.4 Use of part-erected hoists

It is common practice for scaffolders to build three or four lifts of scaffold, raising the components by hand, followed by the hoist company erecting the hoist to the height of the scaffold. Further erection of the part-completed hoist is co-ordinated with the scaffold contractors, so that it can be used for hoisting scaffold components as it follows the scaffold up.

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In such circumstances, normal safe working practices may not be practical. It is important that before any hoist is used to support scaffold erecting, users of the hoists/transport platforms required for transporting scaffolding materials must first ensure that the machine is designated as a lifting platform and thoroughly examined as such before being used for that purpose.



The rules regarding the interlocking of landing gates cannot be applied when some of the gates have not been installed. Detailed guidance is available and it is strongly recommended that anyone faced with this situation on their site obtains and reads it.

The guidance is published jointly by the National Access and Scaffolding Federation and the Construction Plant-hire Association, and is available from both organisations.



For further detailed guidance refer to the Construction Plant-hire Association's *Transporting scaffolding in construction hoists* (Best practice guide).

8.8.2.5 Work at height

Some work at height is unavoidable when carrying out the erection, maintenance, alteration or dismantling of hoists. Work at height must always be planned in accordance with the hierarchy set out in the Work at Height Regulations 2005.



For further detailed guidance refer to the Construction Plant-hire Association's *Work at height on construction hoists* (Best practice guide).

8.8.2.6 Alteration

Alteration of hoists (such as increasing or decreasing the height) should be planned and carried out in the same way as erection.

8.8.2.7 Dismantling

Dismantling operations should be planned and carried out in the same way as erection. Care should be taken to ensure that adequate time is allowed in the construction programme to ensure that dismantling can be carried out without time pressures on the dismantling team. Before dismantling starts it is essential to ensure that the hoist is in a satisfactory condition for safe dismantling.

8.8.3 Safety of hoistways, platforms and cages

Where necessary to prevent injury, hoistways and/or the hoisting machinery must be segregated by a substantial enclosure at ground level. Suitable barriers must be installed at all other access points, over the full height of travel, and wherever persons could be struck by any moving part, to prevent injury and/or falls. Consideration must be given to the area around the base of the hoist with regard to loading and unloading materials, and whether a ramp or pit is required to facilitate access to the cage or platform. The area between the cage or platform and the hoist structure (the threshold) must be suitably protected to prevent any person or material falling through the gap at each landing.

8.8.3.1 Landings and gates

Where access to a construction hoist is required at several levels, consideration must be given to the space requirements at each landing with regard to loading and unloading the hoist. It is essential that each landing and threshold (the area between the platform and landing) can withstand the loads that will be imposed, with particular emphasis on items, such as pallet trucks, which can impose significant point loading. The mechanical or electrical features of the hoist should ensure that the gates at any landing point cannot be opened unless the cage or platform is at that landing point, and the cage or platform cannot be set in motion unless all of the landing gates are closed and locked.



For further detailed guidance on landing and gating requirements for goods hoists refer to the Construction Plant-hire Association's *Safeguarding requirements for landing gates on goods-only hoists* (Best practice guide).

8.8.3.2 Handover

Once erected, the hoist will be subjected to a thorough examination, after which the supplier's competent (appointed) person should arrange to formally hand over the hoist to the user's competent person (lift planner). The user's competent person (lift planner) should arrange for all trained operators to be present at the handover of the hoist to receive:

- familiarisation training on the use of the hoist in normal operations
- instruction on what to do in emergency situations
- instruction on how to carry out the daily, pre-use inspections and weekly inspections.

A handover report, containing details of the instruction given, should be passed to the user's competent (appointed) person.



For detailed guidance on the planning and installation of construction hoists refer to the *Code of Practice for the safe use of construction hoists* (BS 7212).

8.8.4 Planning and use of construction hoists for lifting operations

The planning and carrying out of lifting operations with construction hoists should be in accordance with the principles set out in Chapter C07. In addition, the points below should be taken into account.

8.8.4.1 Operation of hoists

It is essential that only operators who have been trained in the use of the hoist are allowed to operate it. The operator should:

- know the rated capacity (SWL) of the hoist
- be able to accurately assess the weight and distribution of any load brought on to the hoist, and therefore identify if it is overloaded
- have access to accurate wind-speed figures and be aware of any limitations placed on the hoist
- be able to lower the hoist to the next landing in the event of a power failure (passenger carrying hoists only)
- carry out daily, pre-use checks and weekly inspections, and take the appropriate follow-up action as necessary. On some complex hoists, it may be necessary for the weekly inspection to be carried out by an employee of the hoist company.

Each hoist should only be capable of being operated from one position at any one time. Where, as in the case of a goods only hoist, the operator is not carried on the platform, they must have a clear view at all levels from the operating position or, if they have not, arrangements must be made for signals to be given to them at each level.

Whatever the system of signalling used, it must be distinct and clear to the operator.

It is reasonable to expect that the hoist operator and the person(s) giving signals are at least 18 years of age. It is not illegal for younger people to do this work but, under regulations, a specific risk assessment would need to be carried out to ensure that immaturity had been considered. Irrespective of their age, hoist operators must be trained and competent, unless under constant supervision by a competent person for the purpose of training. Competence can be evidenced via the relevant scheme (for example, the Construction Plant Competence Scheme (CPCS Hoist) operator card).

8.8.4.2 Safety notices

The platform of a goods hoist must carry a notice stating:

- the rated capacity (SWL)
- any restrictions regarding load position and concentration.
- that passengers must not ride on the hoist

Cages for passenger hoists must carry a notice stating:

- the rated capacity (SWL)
- any restrictions regarding load position and concentration.
- the maximum number of passengers that can be carried

Landing gates for goods only hoists must carry a notice stating:

- 'Goods only hoist'
- 'Keep gates closed'
- 'No passengers'
- the rated capacity (SWL) in kg.

Landing gates for passenger/goods hoists must carry a notice stating:

- 'Keep gates closed'
- the rated capacity (SWL) in kg.

8.8.5 Carriage of persons in hoists

No person should be carried in a hoist, unless it is a designated passenger/goods hoist or transport platform and it is provided with the following.

- Gates that shut to prevent persons falling out or being trapped between the cage and any other part.
- An efficient interlocking device, which ensures that gates can only be operated when the cage is at the landing place, and that the cage cannot be moved until the gate is closed.
- An efficient automatic overrun device to ensure the cage will come to rest at its lowest and highest points of travel.

The construction of the cage or platform must be such as to protect passengers from falling objects. No person should be allowed to travel in a hoist that is designated a goods only hoist.

8.8.6 Security of loads

All loads must be secured to prevent any part from slipping and falling. Loose materials (for example, bricks and slates) must be lifted in a properly designed box or cage. The platform of a goods only hoist should have sides with a minimum height of 0.6 m, or 1.1 m high if persons can access the platform during loading or unloading.



For detailed guidance on the planning and use of construction hoists refer to the Code of Practice for the safe use of construction hoists (BS 7212).

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8.8.7 Maintenance, checks, inspections and thorough examinations of construction hoists



For further information refer to Appendices A and B.



For further detailed guidance refer to the Construction Plant-hire Association's *Maintenance, inspection and thorough examination of construction hoists* (Best practice guide).

8.8.8 Specific requirements for scaffold hoists, trestle/beam hoists, inclined hoists and gin wheels

Scaffold hoists, trestle/beam hoists and inclined hoists are now commonly used throughout the construction industry. These can be easily transported to the site, are quickly erected and require little operating space. They may be fitted with buckets, skips, platforms or cages. Different models allow for basic height variations of between 8 m and 30 m, although much greater heights can be reached with extensions. Load capacity varies with the model but lifting capacity of 500 kg is not uncommon. This equipment all comes within the scope of LOLER and, although free standing, is subject to the same provisions as construction hoists that are tied to a structure.

8.8.8.1 Scaffold cranes or barrow hoists

These are lifting equipment within the meaning of LOLER and must not be overloaded. The attachment of these to scaffolds is critical, and must be carried out in accordance with manufacturer instructions. They must only be attached to scaffolds made to take the loads imposed.

8.8.8.2 Trestle/beam hoists

These are classed as lifting equipment as defined by LOLER. They rely on counterweighting for stability, at the rear of the trestle frame, and supply of adequate counterweights is essential, often provided by filling bins attached to the frame with sand or gravel. Before use, checks should be made to ensure that sufficient material has been added to bins, and that they are in good condition so that ballast cannot leak out.



For further information on maintenance, checks, inspections and thorough examinations of these two hoist types, refer to Appendices A and B.

8.8.8.3 Inclined hoists

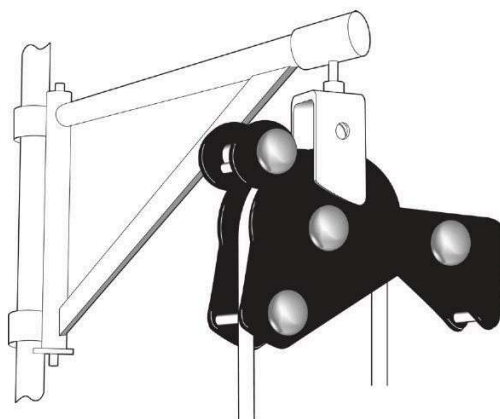
These are lifting equipment as defined by LOLER. They are particularly useful for trades such as bricklaying and traditional roofing, and have the potential to eliminate a significant amount of manual handling. Inclined hoists are also known as roof tile or brick carriers.

New developments include telescopic and slewing functions. Load capacities go up to 250 kg and lateral extensions may be up to 35 m. They are specifically designed for use at angles between 10° and 85°, and may incorporate a hinged section to allow the hoist to follow the pitch of a roof. They provide flexible solutions to the problems of removing demolition waste from locations at height, utilising a conveyor-belt system, so care needs to be taken to ensure that the guards to the rollers are in place to ensure that fingers and/or clothing cannot be dragged in. Maintenance, inspection and thorough examination should follow the requirements for construction hoists.

8.8.8.4 Gin wheels

In many cases provision has to be made to raise tools and light materials manually with a rope and gin wheel or single block. Whilst these are still commonly used, modern variants have an automatic locking mechanism (inertia braked) to prevent the load from going into free-fall if the hoisting rope is accidentally released. The following requirements apply when connected to a scaffold.

- Any scaffold tubes and hooks should be strong enough to take the load, and be properly secured to prevent movement.
- All ropes should comply with the relevant BS EN standards and fit the wheel correctly. They should be marked with a tag confirming their rated capacity (SWL).
- Preferably, the gin wheel should be suspended from a ring-type fitting passed over the end of the supporting tube and secured against lateral movement by scaffold fittings. If using a hook type, it should be properly lashed and moused.
- Any joints in standards should be made with sleeve couplers.
- Gin wheels should be suspended no more than 750 mm from the outer scaffold standard.
- Hooks used for supporting materials should be safety hooks and spliced into the rope.
- The maximum load should be no more than 50 kg at 750 mm from the outer scaffold standard.



An inertia braked gin wheel utilising a purpose built bracket system. Inertia braked wheels are now more common and far safer than traditional gin wheels

Whilst of simple construction, gin wheels are categorised as lifting equipment, so must be thoroughly examined before use and then at regular intervals, and the results recorded. As a guide, once weekly should be sufficient. Similarly, lifting accessories (such as the rope, hook and shackles) must be thoroughly examined before use and then at least every six months. Given the scope for the harsh treatment of this type of equipment, all parts of a gin wheel assembly should be thoroughly examined at least every six months. The extent of any thorough examination should reflect the risks that would arise from its failure.

8.9 Mast climbing work platforms (MCWPs)

A mast climbing work platform (MCWP) is a form of temporary working platform giving variable height access to structures. They allow work to be carried out on the façade of buildings safely, efficiently and economically. MCWPs are commonly used during window replacement, lift shaft installation, curtain walling, bricklaying, cleaning and refurbishment.

They comprise a powered platform attached to one or more vertical masts, with mechanical drive units on the masts. Two or more units have to co-ordinate together to get to the required height.



For further information on MCWPs refer to Chapter D02 Working at height.

8.9.1 Training

Training in the safe use of a MCWP on sites is available for mobile operators, demonstrators, installers and advanced installers. In addition, CPCS offers a category for mast climbing. Training is differentiated according to the following job categories and responsibilities.

Demonstrators are responsible for inducting users and ensuring that the equipment is used safely. They need to understand model-specific servicing and engineering details.

Installers, who install the platforms, must have detailed understanding and experience in erection and dismantling procedures for the models that they will be installing or dismantling.

Advanced installers are trained as managers or supervisors. They will have the ability to plan, carry out a risk assessment and method statement, and to configure special arrangements, including special tie and anchor specifications.



Health and Safety Executive – Safety alert

The industry is alerted to a serious technical fault that the HSE has found with MCWPs used widely across construction projects.

The problem

Following investigation, it has been discovered that some MCWPs that rely on two independent motor drive units per mast as the means to prevent falling with overspeed are not fitted with suitable and sufficient controls to manage this risk. This means platforms could fall from height where mechanical faults in drive units go undetected. If not rectified quickly, the fault could cause serious injury, or even death.

Failures in drive units can be such that neither the centrifugal brakes (intended to limit the speed of descent) nor the automatic brakes (intended to engage when powered travel is stopped) are able to have an effect.

Immediate action required

The installer must take immediate action as required, and check that necessary control measures are in place for all MCWPs in use or available for use on site. If measures are not in place, they need to withdraw MCWPs from use until those responsible for supply, installation, inspection, servicing, maintenance and for thorough examination of MCWPs ensure that the following are carried out.

- There is the means to identify a loss of mechanical integrity in each drive unit to prevent falling with overspeed.
- Each unit is fitted with a mechanical device (e.g. centrifugal brake) to automatically prevent an excessively quick descent.
- Damage to drive units due to platforms being powered onto buffers/base frames is prevented.
- Platforms and associated equipment are not damaged by physical overloading.
- Thorough examinations, inspections and tests, and visual and functional checks are appropriately planned and carried out.

For further information refer to:

- **HSE – Safety alert – Mast climbing work platforms: Failure to detect mechanical failure in drive units leading to uncontrolled fall of platforms**



- **Lifting platforms – Mast climbing work platforms (BS EN 1495)**
- **Code of Practice for the installation, maintenance, thorough examination and safe use of mast climbing work platforms (MCWPs) (BS 7981)**
- **The Construction Plant-hire Association's Guidance on tying construction hoists and mast climbing work platforms to supporting structures.**

LIFTING EQUIPMENT

8.10 Telehandlers (telescopic handlers)

Telehandlers (also known as rough terrain telescopic handlers or variable reach trucks) make a valuable contribution to the construction process by enabling materials to be unloaded from delivery vehicles, transported around construction sites and placed at height (reach).

They are versatile machines that can be fitted with a wide range of attachments (such as buckets, skips, work platforms and crane jibs). The manufacturer's recommendations for the fitting and use of attachments should be followed. Unfortunately, there have been a significant number of accidents involving the use of telehandlers, which have tragically included a number of fatalities.

As with all lifting equipment the safe use of telehandlers requires selecting the right equipment, which has been adequately maintained, planning the operation effectively and having a trained and competent operator.

8.10.1 Types of telehandlers

Telehandlers fall into two broad categories, non-rotating and rotating.

8.10.1.1 Non-rotating

These machines comprise a powered wheeled chassis onto which is mounted a telescopic boom, pivoted on the chassis, which can be elevated from below the horizontal to an angle approaching the vertical.

The outer end of the boom is fitted with a fork carriage and forks for handling unit loads. Levelling of the forks in the longitudinal plane, as the boom elevation changes, is carried out automatically.

These machines are able to transport loads from one part of a site to another and place the load at height. On construction sites, the wheels are generally fitted with lug grip tyres to enable the chassis to negotiate rough uneven ground.

Various drive and steering configurations are available (such as two-wheeled drive, four-wheeled drive, pivot steer, rear-wheel steering and four-wheeled steering).

Many telehandlers are fitted with stabilisers that are deployed when the machine is stationary to provide additional stability and enhance the machine's lifting capacity.

Telehandlers are often fitted with a feature that allows the chassis to be levelled laterally, where appropriate, before the boom is raised, when the machine is standing on uneven ground.

8.10.1.2 Rotating

Rotating telehandlers have all of the features of the non-rotating type, but with the addition of a rotating or slewing superstructure on which the boom and operator's cab are mounted.

These machines also have outriggers fitted at either end of the chassis that enable the entire chassis to be lifted clear of the ground for maximum stability.

The main advantages of these machines over the non-rotating type is compact chassis size, enhanced lifting height, increased stability and ease of placing loads without moving the chassis.

8.10.2 Telehandler attachments

Telehandlers are versatile machines which, in addition to the lifting of unit loads on forks, can be fitted with a wide range of attachments. Some examples are shown below.

- Sideshift forks.
- Sweepers.
- Block grabs.
- Tipping skips.
- Crane hooks.
- Crane jibs.
- Buckets (general purpose and material handling).
- Integrated access platforms.

It is essential that all attachments are compatible with the telehandler with which they are to be used.

Where necessary, the telehandler manufacturer should be consulted where third-party attachments are to be used.

Planners and operators should be aware that attachments will alter the rated capacity (SWL) and centre of gravity of the machine.



Typical non-rotating telehandler



Typical rotating telehandler

8.10.3 Quick hitches

Some telehandlers are fitted with quick hitches or quick couplers, which enable attachments to be changed easily and rapidly. They fall into two types.

Mechanical quick hitch, where the hitch is engaged with the attachment, using the boom functions combined with fork carriage tilt. Once the quick hitch and attachment are engaged, a locking pin is inserted and secured with a retaining pin.

Hydraulic quick hitch, which is engaged in the same manner as the mechanical quick hitch but the locking pin is engaged hydraulically using the controls in the telehandler cab.

Both types of quick hitch can allow the attachment to become detached from the quick hitch if the manual locking pin is left out or the hydraulic locking pin fails to engage fully. There have been several serious injuries caused by falling attachments and misuse.



It is essential that operators get out of the cab to physically ensure that all quick hitches are securely locked before starting work with a newly fixed attachment.

For further detailed guidance on quick hitches refer to the:



- **Strategic Forum for Construction Plant Safety Group's *Safe use of quick hitches on excavators* (Best practice guide)**
- **HSE's online construction safety topic *Industry guidance on quick hitches*.**

8.10.4 Planning and use of telehandlers

The planning and carrying out of lifting operations with telehandlers should be in accordance with the principles set out in Chapter C07. In addition, the points below should be taken into account.

8.10.4.1 Visibility

Restricted visibility when the boom is raised or when large loads are carried, as well as poor pedestrian segregation, have been identified as major causes of accidents involving pedestrians and telehandlers. Telehandlers are often fitted with aids to improve visibility and the operator's awareness of people in the vicinity of the telehandler.

These aids should be in good working order and properly adjusted. It is the operator's responsibility to check the condition of all secondary aids to visibility and **not to use the machine** if they are not present or not working correctly. It is the site manager's or supervisor's responsibility to fully support the operator in this action.

Some telehandler operations present particular challenges. A suspended load, for example, will clearly block some forward vision, but the raised boom may also obscure vision to the side. If a telehandler is to lift suspended loads on site, the physical dimensions of the loads and their effect upon visibility must form part of the risk assessment.

If a telehandler is used to load or unload a truck, the partially raised boom will obstruct vision to the forward offside and may block the wing mirror that gives visibility to the rear offside. The safest way of using a telehandler to load/off-load is to keep the machine stationary and use the telescopic facility of the boom, rather than using the wheels. The truck driver should remain in a designated safe location.

Whilst the principal contractor has the primary responsibility to ensure adequate segregation and the supervisor must ensure that it is enforced, it remains the operator's responsibility to look around and check for the absence of pedestrians before moving and whilst manoeuvring and travelling. If the operator cannot see clearly, they should seek assistance or leave the cab to look around to confirm it is safe to continue the procedure.

8.10.4.2 Travelling on inclines, slopes and gradients

The telehandler must only operate on slopes or gradients at a speed designated by the site, taking into account ground conditions, to ensure that the operator has full control at all times. Traffic routes should be on consolidated ground or temporary roadways giving equivalent safety.



Consolidated ground is firm ground capable of accepting the mass of a loaded telehandler without significant deformation.

The slopes or gradients given in the table are to assist planners in the preparation of traffic routes. They are the maximum gradients on which a telehandler can travel while being operated when crossing sites **in the standard travelling mode and at walking speed**.

The standard travelling mode of a telehandler is with the boom retracted, the load/fork arms lowered to provide a clearance of 300 mm and not more than 500 mm (see manufacturer's instruction manual) from the ground to the upper faces of the fork arms, and the fork arms fully tilted rearwards. Operation on any traffic route with a greater slope or in a travelling mode different to the above should be subject to a site-specific risk assessment.

Slope	%	Gradient	Angle
Maximum down slope	15%	1 in 6.66	8.5°
Maximum up slope			
Maximum lateral slope			

LIFTING EQUIPMENT

8.10.4.3 Stability of the telehandler

The stability of telehandlers and loads are affected by the condition of the ground on which they travel or stand when loading or unloading.

The area selected by the planners as a static loading area must be large enough to accommodate all the wheels of the telehandler and stabilisers or outriggers when fitted. The area should be of consolidated firm ground or surfaces giving similar levels of safety. It should be capable of accepting the mass of the loaded telehandler without significant deformation and be substantially level in both planes to ensure lateral and longitudinal stability when lifting operations are being carried out. Substantially level ground is defined as ground with a gradient of ideally 1% (1 in 100 gradient, 0.6°) but not more than the standard drainage slope of 2.5% (1 in 40 gradient, 1.4°).

When determining the area to be used for the unloading of lorries and the storage of materials, care should be taken to ensure the ground is consolidated and level. The area provided must be large enough that the telehandler will not need to make tight turns with an elevated load. A telehandler may be used for loading/unloading operations in areas that are not level if they are used within their design capabilities. Where the achievement of a level loading or unloading area is not reasonably practicable, a risk assessment will be necessary.

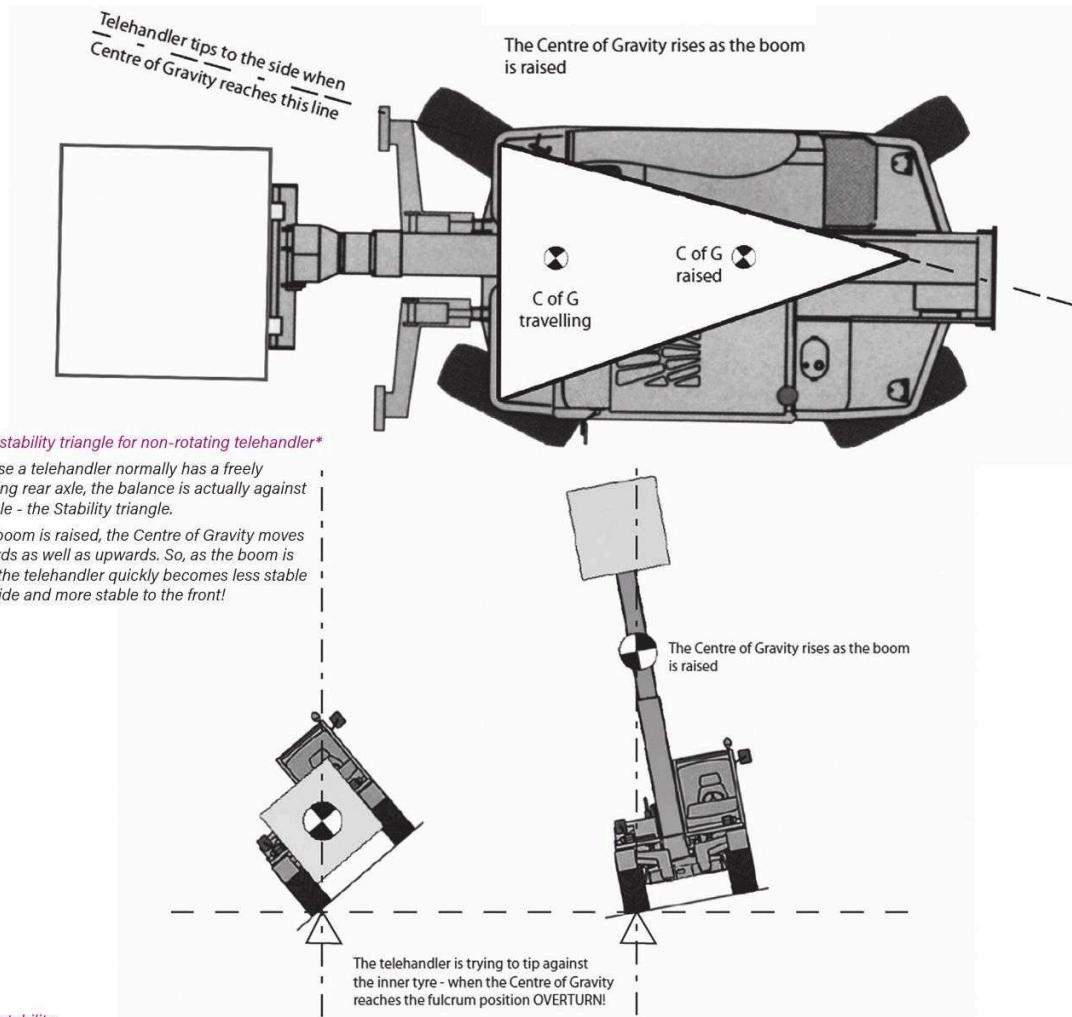


Standard telehandlers are normally fitted with load movement indicators (LMI) and, in some cases, load movement limiters. These prevent components (such as the boom) from extending if an overload situation is indicated. The operator should never rely on the LMI as a way of checking whether the machine is safe to pick up a load. Refer to the lift plan and lift capacity charts of the machine at all times.



Load movement indicator units only warn of longitudinal instability. They can become inaccurate and require resetting by a competent (appointed) person on a regular basis.

Telehandlers become less stable in a lateral or sideways direction as the boom and load are raised and, because telehandlers normally have freely oscillating rear axles, the tipping lines form a triangle, rather than the rectangle of machines supported on outriggers (such as rotating boom telehandlers and mobile cranes). These effects are shown in the following diagrams.



Lateral stability

! When travelling with a load, the load should always be kept as near to the ground as possible.

Tyres play a vital part in the stability of telehandlers. Stability and load carrying capacity can be adversely affected by issues such as those shown below.

- Incorrect tyre pressures.
- Incorrect ply rating.
- Poor tyre repairs.
- Differences in diameter of tyres on the same axle due to differential wear.
- Tyres with identical nominal sizes being of different physical size.
- Use of tyres that are not of the required performance specification.

8.10.4.4 Wearing of seat belts

Telehandler cabs are provided with seat belts to restrain the operator and reduce the risk of serious injury in the event of a machine overturning. Machines should be fitted with green flashing lights indicating that the seat belt has been activated. A survey of telehandler accidents in the UK carried out by the HSE shows that in seven years there have been at least 72 lateral overturns of telehandlers.

In three cases, the operator was killed by being ejected from the cab and crushed; in a further two cases the operator was ejected from the cab and received serious crushing injuries. Had they been wearing seat belts, their deaths or injuries would probably have been avoided.

! It is vital that operators of telehandlers wear their seat belt at all times – it could save their lives.

8.10.4.5 Stability of the load

When using a telehandler for loading or unloading a vehicle or stack, care must be taken to ensure that the load on the vehicle or stack remains stable. Unless the correct sequence of loading or unloading is followed there is risk of the load becoming unstable with the potential for injury to persons in the vicinity.

- Loading should always be carried out with unloading in mind.
- Vehicles should always be loaded as directed by the vehicle driver, working from side to side and distributing the load evenly.
- Unloading should be carried out by reversing this sequence.
- When loading uncoupled articulated trailers, the first part of the load should be placed over the rear axle before loading from the front of the trailer.
- Care should also be taken to ensure that the part of the load on the vehicle or stack is not dislodged by contact with the telehandler carriage, forks or other attachments.
- During loading and unloading, measures (such as exclusion zones or barriers) should be put in place to ensure that personnel are kept clear as there is a greater potential for falling loads.
- Before lifting a load, it should be assessed to ensure that it will be stable and secure during lifting.
- The forks should always be evenly spaced on either side of the centre of gravity of the load.
- Long or irregularly shaped loads may have a centre of gravity that is close to the edges of the forks and may well result in the load falling during lifting.
- Long loads can be easier to control using a wider fork carriage. Personnel should be kept clear of the load and should **not** be asked to steady the load.



Use of a wider fork carriage

! The practice of undercutting, where the load is lifted on the ends of the forks, rather than back at the heel, closest to the fork carriage, is often used when the load is to be placed adjacent to a wall or another item. This practice will reduce the rated capacity of the telehandler and may affect the stability of the load as its centre of gravity may be near or beyond the end of the forks.

8.10.4.6 Use of lifting hooks or crane jibs for suspended loads

Lifting and travelling with suspended loads is not the primary purpose of telehandlers. When selecting equipment for this activity, first ensure that a telehandler is suitable for carrying out the task safely. Where other equipment is more suitable, it should be used.

LIFTING EQUIPMENT

Using a telehandler to lift and travel with a suspended load is generally more hazardous than lifting unit loads on the forks of a telehandler. The operation should be effectively planned and communicated to all those involved (*for further information refer to Planning of lifting operations in Chapter C07*). The lifting of suspended loads should always be carried out with a telehandler that is fitted with a suitable lifting hook or attachment. If this can be shown to be not reasonably practicable, it is essential that any other method is justified by a robust risk assessment, taking account of the hierarchy of control.

Any lifting hook or crane jib should be marked with a rated capacity (SWL) that must not be exceeded; the rated capacity of the hook may be less than the rated capacity of the telehandler or vice versa. It is important that the lower of the two values is used to determine the rated capacity of the combination. When working out the total weight of the load to be lifted, the weight of the lifting hook or crane jib and any lifting accessories (shackles, slings, and so on) must be taken into account.

Telehandler manufacturers supplying lifting hooks and crane jibs for their own machines will generally provide reduced rated capacities where suspended loads are being lifted and carried. This is intended to reduce the risk of overload or overturn due to displacement of the load through travelling, boom movements or the effect of wind.

Extreme care should be taken when travelling with a suspended load, as any movement of the load will alter the load radius and may affect the stability of the telehandler. The signaller's position should also be considered, as they are at risk of being in close proximity.



Basic telehandler operator training does not include lifting or travelling with suspended loads

If these tasks are to be carried out, the employer must ensure that the operator is suitably trained and assessed as competent. The majority of basic training courses only deal with fork-mounted unit loads. Only training providers who can demonstrate having appropriate expertise of suspended loads and lifting operations with telehandlers or mobile cranes should be selected. Any training should also include an appropriate element of slinging and signalling.



For further guidance on suspended loads refer to the Strategic Forum for Construction Plant Safety Group's *Lifting and travelling with suspended loads using telehandlers* (Good practice guide).

8.10.4.7 Use of telehandlers for the lifting of persons

Telehandlers can be used with working platforms to allow people to work at height. It is generally accepted that an integrated working platform provides a higher level of safety than a non-integrated type and can be used for the lifting of persons where a MEWP is not available.



Non-integrated platform attachments must not be used for tasks on construction sites.

8.10.5 Maintenance, checks and inspections of telehandlers



For further information on maintenance, checks and inspections refer to Appendix A.

In addition, the points below should be taken into account.

8.10.5.1 Tyre repair, replacement and maintenance

This section applies to pneumatic tyres only. Foam-filled tyres are not user serviceable.

Tyre pressure check

- Tyre pressures should be marked on the telehandler chassis adjacent to each wheel.
- Pressures should be checked daily when the tyres are cold.
- Tyre pressures should be the same for all tyres on the telehandler and should be within +5% and -0% of the manufacturer's specified pressure, unless a different tolerance is specified by the manufacturer.
Note: research undertaken by a major telehandler manufacturer indicates that -5% reduction in tyre pressure below the manufacturer's specified value can result in a 30% reduction in rated capacity at certain points on the load/height/radius curve.
- Tyre pressure gauges should be of known accuracy (within +/- 1%).
- Where a tyre is found to be significantly under pressure, no attempt should be made to reinflate it and return the machine to use. It is essential that the cause is investigated by a competent tyre specialist, as this may be an indication of imminent tyre failure.

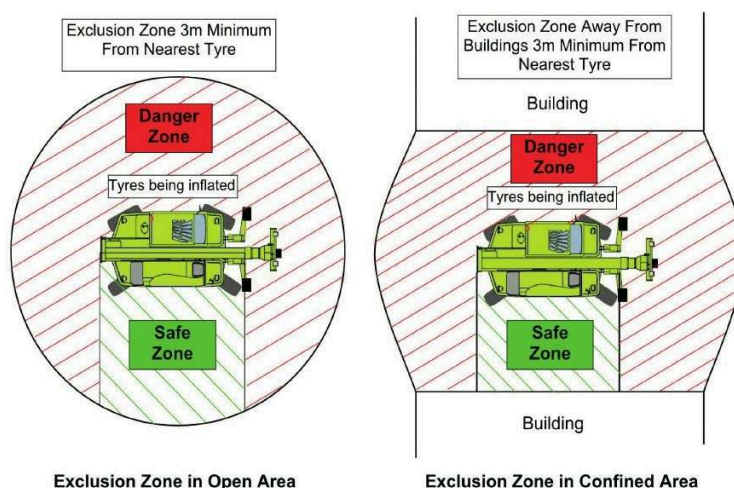
Tyre inflation applies to tyres fitted on single-piece wheels. Where multi-piece wheels are fitted or there is any doubt about the type of wheel, a competent tyre specialist must be consulted.



For tyres fitted on multi-piece wheels refer to the BITA publication *Multi-piece pneumatic tyred wheels - Inspection, maintenance and repairs* (GN67).

- Provided the tyre has been checked and confirmed to be safe to inflate, air should be added, if required, inflating to the pressure specified in the manufacturer's manual.

- Before starting inflation a personnel exclusion zone of at least 3 m around the machine should be established by the use of barriers, cones, and so on (*refer to image below*).
- Personnel inflating tyres should stand a minimum of 3 m away from the tyre and outside the likely explosion trajectory to avoid injury in the event of a failure. This will require at least 3 m of airline between the nozzle and the airline trigger mechanism (*refer to image below*).
- Personnel should ensure that they stand on the other side of the telehandler when inflating (*refer to image below*).
- Tyre valves should be checked to ensure that they are not leaking. Valve stem caps should always be replaced.



Tyre inflation safe zones and exclusion zones

www For further detailed guidance on the safe inflation of tyres refer to the HSE publication *Safety during tyre inflation in motor vehicle repair* (INDG433).

e.g. A real-life example of tyre issues

A telehandler operator had been checking the tyre pressures on his machine. Having completed the checks, he was walking away from the machine when one of the tyres exploded. Subsequent investigation revealed that the tyre, which had recently been replaced, was 14 ply with a 3.5 tonne load rating rather than the manufacturer's specification of 16 ply with a 6 tonne rating. The contractor also found that the tyre had been ordered from an approved supplier by asking for a tyre for that model of telehandler, without any mention of ply or load rating. A subsequent check of other machines found that 30% were fitted with incorrect tyres.

Tyre damage. All tyres should be inspected daily. The tread and side walls should be checked for bulges, separation and cuts. For pneumatic tyres, when any cut, rip or tear is discovered that exposes sidewall or tread area cords in the tyre, measures must be taken to remove the product from service immediately. Arrangements must be made for replacement of the tyre or tyre assembly.

Tyre replacement. Telehandler manufacturers generally list a range of approved tyres in the parts manual for a specific machine. If not using an approved replacement tyre, consult with a competent tyre specialist and ensure replacement tyres have the following characteristics.

- Same physical size as the original.
- Tyre tread contact width equal or greater than original.
- Equal or greater ply and load rating than the original.
- Equivalent performance specification.

Unless specifically approved by the manufacturer, foam-filled or ballast-filled tyre assemblies must not be replaced with pneumatic tyres. When a tyre is being replaced it is essential that the condition of the other tyres is assessed by a competent tyre specialist and appropriate action taken. This may include moving partly worn tyres around, to ensure that the telehandler is substantially level when standing on a flat, level surface. If tyres on opposite sides are different sizes, the telehandler boom will not be vertical when the machine is standing on level ground. This will cause the combined centre of gravity of the telehandler and load to move sideways, which may lead to instability.

8.10.5.2 Wheel replacement

It is essential that replacement wheels have the same diameter, width and offset dimensions as the originals.

8.10.6 Thorough examination of telehandlers

Q For further information on thorough examination refer to Appendix B.

www For further information refer to the Strategic Forum for Construction Plant Safety Group's *Safe use of telehandlers in construction* (Good practice guide).

8.11 Excavators used for lifting

The use of excavators for lifting operations, particularly on construction sites, has become more common over the last few years. Excavators and backhoes are primarily designed for earthmoving, not for lifting operations. When planning a lifting operation, consideration should be given as to whether an excavator is the most appropriate machine, taking into account the type of lift and the duration of the task.

The use of an excavator or backhoe for lifting creates additional hazards for personnel in the vicinity. Under normal circumstances, personnel are kept away from the working area around the bucket, as this is considered to be a hazardous area. Where the excavator is used for lifting, the slinger has to be in the danger area in order to hook the load onto the hooking device. This puts the slinger at risk of being struck by the load, bucket or excavator arm if the excavator moves without warning.

8.11.1 Planning and use of excavators used as cranes

The planning and carrying out of lifting operations with excavators should be in accordance with the principles set out in Chapter C07. In addition the following points should be taken into account.

8.11.1.1 Excavator requirements when used for lifting

The design of earthmoving machinery for lifting (object handling) is covered by BS EN 474 Parts 1, 3, 4 and 5. Any such machine designed for object handling should have a rated object-handling capacity table available inside the cab. If this is not available, then the machine should not be used for object handling.



Handling attachments (for example, grab) that do not require the assistance of a person for hooking or guiding are considered to be part of normal earthmoving operations and do not require warning devices and a rated capacity (SWL) table.

An earthmoving machine used for lifting operations must be fitted with a load hooking device. If the device is a hook, then this should have a clip or other device that prevents a sling slipping off the hook. Many quick hitches provide a load hooking device in the hitch design. If the rated lifting capacity for an excavator or the backhoe portion of a backhoe/loader is greater than one tonne (or the overturning moment is greater than 40,000 Nm) then the machine must be fitted with the following.

- A boom-lowering control device on the raising boom cylinder(s) that meets the requirements of ISO 8643:2017.
- An acoustic/visual warning device that indicates when the object-handling capacity or corresponding load moment is reached.

A change in the definition of rated object-handling capacity, in the latest version of EN 474, affects excavators manufactured and CE marked after 29 December 2009 and UKCA marked on any equipment purchased from 1st January 2021. It requires any new excavator that has a rated capacity (SWL) of over one tonne, and which is going to be used for lifting, to be fitted with a rated capacity indicator (RCI) and check valves. Previously, some manufacturers had been marking the excavator for a rated **lifting capacity** of one tonne even though it had a **rated capacity** in excess of this. Excavators manufactured and CE marked before 29 December 2009 are not affected by these changes.



- **Loaders and the loader portion of a backhoe/loader do not require a boom-lowering control device or acoustic/visual warning devices.**
- **Where a risk assessment shows that there is a significant risk of overloading and/or overturning on machines with a rated capacity (SWL) of one tonne or less, a rated capacity indicator may be required (refer to LOLER ACoP and Guidance).**

Suitable slings must be available to attach the load to the excavator. Slings and other lifting accessories should be CE marked, or UKCA marked on any equipment purchased from 1st January 2021, and marked with the rated capacity (SWL).

Whilst BS 7121 (*Code of Practice for safe use of cranes*) may not specifically refer to excavators used as cranes, compliance with all the appropriate parts of BS 7121 would assist in the provision of safe systems of work, as required by Section 2(2) of the Health and Safety at Work etc. Act 1974.

8.11.1.2 Lifting of persons with excavators

Excavators should not be used under any circumstances for the lifting of persons, as they are primarily designed for excavating with a bucket and consequently are capable of operating speeds and movements that make them totally unsuitable for the lifting of persons.

8.11.1.3 Load attachment

Where the hooking device (the point on the machine designed for connection of the load) is not part of the bucket, the bucket should (where possible, and unless the manufacturer's instructions specify otherwise) be removed in order to improve visibility and reduce the weight being lifted. If the bucket is retained, then the weight of both the bucket and quick hitch has to be added to the load when determining whether the load is within the rated capacity.



When attaching lifting slings to the hooking device or lifting point care should be taken to ensure that the slings and their attachments are able to hang free at all times.



Attachment using a shackle may limit rotation if, for example, a pipe suspended from the slings is to be turned end to end



Hitch tilted backwards with master link subject to bending



If the quick hitch is tilted backwards and/or the dipper arm is raised, the master link of the sling and any attachments may bend or twist – possibly leading to damage or failure.



Hitch tilted backwards with master link subject to twisting



Using a swivel to overcome this problem



Good practice of chain and master link hanging freely without obstruction



For further information on quick hitches refer to *Quick hitches* within Chapter C06 Mobile work equipment.



It is essential that operators get out of the cab to physically ensure that all quick hitches are securely locked before starting work with a new attachment.



For further guidance on planning and carrying out lifting operations with excavators refer to the Construction Plant-hire Association's *Lifting operations in construction when using excavators* (Guidance).

For further detailed guidance on quick hitches refer to the Strategic Forum for Construction Plant Safety Group's *Safe use of quick hitches on excavators* (Best practice guide).

LIFTING EQUIPMENT

8.11.2 Maintenance, checks, inspections and thorough examination of excavators used as cranes



For further information refer to Appendices A and B.

In addition the following points should be taken into account.

8.11.2.1 Thorough examination of quick hitches

Quick hitches used for lifting fall under the thorough examination requirements of LOLER, whilst those not used for lifting fall under the inspection requirements of PUWER. The periodic examination and/or inspection of quick hitches forms a vital addition to planned preventive maintenance. It provides evidence of the adequacy of a maintenance regime and highlights any deficiencies, leading to improved reliability and safety of the hitch being used. The requirements for examination or inspection vary with the type of hitch, and consequently the type of examination/inspection and its frequency is a decision for the competent (appointed) person carrying out the examination/inspection.



For information on the legal minimum requirements for quick hitches see the current version of the HSE's *Quick hitch devices on excavators 23.10.13*. In all cases, seek guidance from the competent (appointed) person regarding the most appropriate regime for a particular hitch.



Quick hitches

The nature and extent of both LOLER thorough examinations and PUWER inspections are risk based and since the risks associated with quick hitch failure are similar for lifting and non-lifting duties, good practice requires that all quick hitches:

- that remain permanently fitted to a machine are thoroughly examined and/or inspected at the same time as the base machine
- suitable for lifting (fitted with a lifting eye), which are regularly removed from the machine (as interchangeable equipment), are subjected to a periodic thorough examination at intervals not exceeding six months.

All quick hitches subject to thorough examination must be permanently marked with a unique identification number. If a quick hitch does not have such a number, the owner/user must ensure that one is applied. Reports of thorough examination should contain the details required by Schedule 1 of LOLER.



Refer to Appendix C for information on Schedule 1 of LOLER.

Thorough examinations should be carried out by competent (appointed) persons who have the genuine authority and independence to ensure that examinations are properly carried out and that the necessary recommendations arising from them are made impartially.

8.11.2.2 Thorough examination/inspection following major alteration, damage or incident

If the quick hitch has been subjected to major alteration, repair, damage or has been involved in an incident, it should be thoroughly examined/inspected before it is put back into service.



e.g. Major alterations

- Alteration of attachment mechanisms.
- Attachment of lifting eyes.
- Modification of pin bores.
- Application of heat.

8.12 Lifting accessories

Lifting accessories are items of equipment used for attaching the load to the lifting equipment (such as chains, ropes, slings, hooks, spreader-beams, shackles and eye bolts). All lifting accessories must also be:

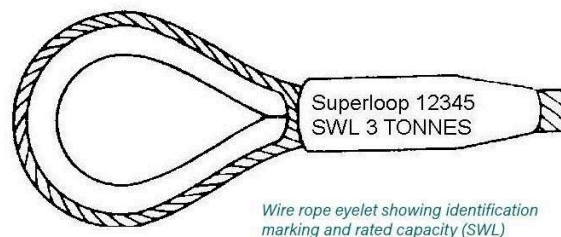
- properly constructed and maintained
- free from any defect or damage likely to affect their strength
- regularly maintained and inspected
- thoroughly examined
- securely attached to the lifting equipment (for example, the crane) and the load
- used within their rated capacity (SWL).

Accessories used for lifting must be thoroughly examined at six monthly intervals, or more regularly if being used in harsh conditions with records kept as specified in Appendix B.

8.12.1 Marking

All lifting accessories must be clearly marked with their rated capacity (SWL) and carry an identifying mark. Where this is not possible then a coding system (such as a tag or colour code) should be used to allow the user to determine the rated capacity (SWL). A lifting accessory may also be marked with its own weight, which is a consideration when assessing the total load to be lifted.

Where the rated capacity (SWL) depends on the configuration of a lifting accessory (such as a two leg chain) then the rated capacity (SWL) for each configuration should be clearly marked on the equipment using a tag or by a chart available at the point of use.



8.12.2 Overloading

Lifting accessories must never be overloaded, except under test conditions authorised by an experienced and competent (appointed) person.

8.12.3 Preventing damage

The edges and corners of a load should be packed to prevent sharp edges damaging lifting ropes, chains or slings.

8.12.4 Hooks

All hooks used for lifting must be fitted with a safety catch, or otherwise shaped to prevent the sling eye or load coming off the hook.

8.12.5 Slings

Slings must be attached correctly to the lifting equipment by an approved method, either by securing the ring directly onto the hook if size permits, or by use of a suitable shackle, fitted with the pin of the shackle on the hook and the load suspended from the bow.

The correct method of slinging will vary with the types of load, the different materials or the items lifted. It is essential to see that the load is secure. Care must be taken to see that slings are not damaged. Suspect or defective slings must be discarded.

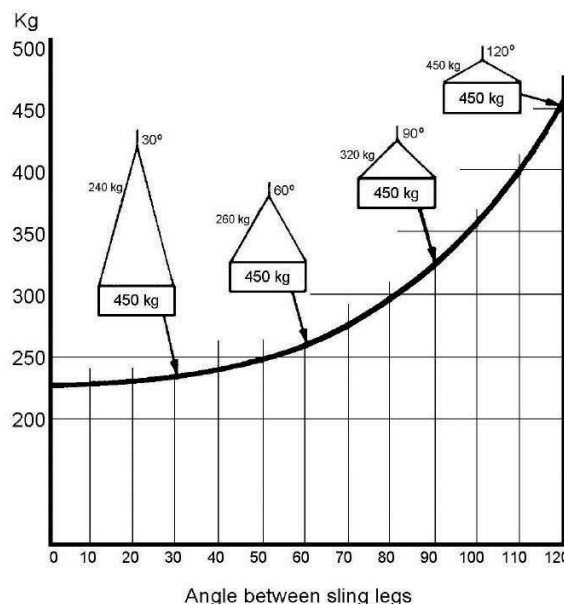
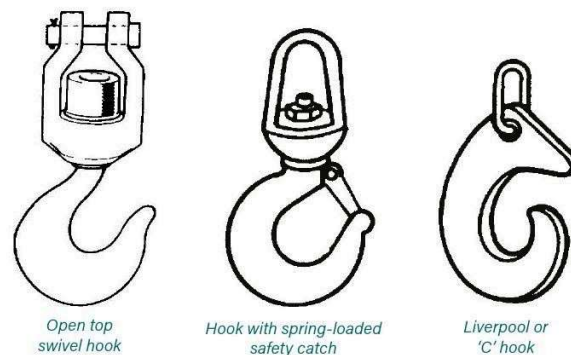
Tag lines, securely attached to the end of the load, should be used when handling long or large loads, to direct the load into position and prevent it spinning. Tag lines should be as short as possible.

Multiple slings (two-legged, three-legged, and so on) must be connected by a ring or shackle and the load properly distributed so that no leg is overloaded. Chandelier lifting (using each leg to lift a load at different heights) must not be undertaken when using multi-legged slings as this would place the slinger in danger when detaching the load.

When the sling is in use, the angles between sling legs should be less than 90°. At angles greater than 90°, the strain on each leg increases rapidly to a point where they may break because of overloading.

With a simple two-legged parallel sling, the load on each leg is half the total load. As the angle between sling legs increases, the load on each sling leg increases to approximately double at an angle of 120°.

With a sling angle of 90°, the rated capacity (SWL) of the sling should be at least 43% greater than the nominal weight of the load.



Good practice showing the load in each sling leg increasing as the angle between the sling legs is increased

8.12.6 Repaired lifting accessories

Any lifting accessory (chain, sling, and so on) that has been altered or repaired by welding must be thoroughly examined and may require testing before being put back into service. The record of thorough examination should be kept until at least the next scheduled thorough examination.

LIFTING EQUIPMENT

Knotted ropes, chains or slings and those that have been shortened or joined by nuts and bolts through the links must not be used.

Any lifting accessory that is thought to have suffered a loss of strength, been damaged or overloaded, or is otherwise defective, should be withdrawn from use and quarantined. The accessories should be disposed of or re-examined by a competent (appointed) person who will recommend that either it can be put back into service or must be disposed of.

8.12.7 Construction of slings

Slings are available in a wide range of styles made from many different materials to suit particular purposes.

Chain slings are made from various grades of steel. They can stretch and the links, rings or hooks may become distorted and fracture if subject to excess stress.

Chain slings should only be shortened by using the correct shortening clutches. Chains must not be knotted or joined by nuts and bolts. Hooks must be of the 'C' type or fitted with a safety latch. No chain sling should be used if the angle from the vertical exceeds 60°, as beyond this point the forces in the legs drastically increase.

Chain slings or lifting chains should **never** be used for towing a site vehicle horizontally.

Wire rope slings are made from drawn steel wire. Each leg of the sling will have an eye formed at either end. Wire rope slings may be damaged when kinked sharply or if put under stress when twisted. Steel wire rope may be damaged by corrosion through poor care and storage.

Wire rope slings must not be made up on site using bulldog grips.

Wire ropes and slings are supplied in many different types of construction, for different lifting applications. It is therefore important, when ordering an item, to specify the intended use.

Fibre rope slings might be made from natural fibres (manila, sisal or hemp) or synthetic fibres. Slings made from natural fibres can be prone to rotting. Only purpose-made slings, clearly marked with their rated capacity (SWL), should be used on site.

On no account should slings be fabricated from lengths of rope found lying around the site. Fibre rope slings are more easily cut or damaged, and should be visually examined by a competent person every time before use to ensure they are serviceable. Natural fibre ropes should not be used for making up slings on site.

Synthetic fibre ropes do not rot but can be affected by some chemicals. Care should be taken to avoid contamination with alkalis or acids.

Suspect or contaminated synthetic fibre ropes and slings must not be used.

Flat lifting slings and stops are used where special lifting operations are required and afford a certain amount of protection to the load.

Flat and round slings may be made of woven synthetic materials (such as nylon, polyester, polypropylene and Terylene) with 'eyes' sewn in, or plastic-coated wire mesh, or formed by a series of plaited wire ropes (rope made from strands of high tensile wire) between two end fittings. These might also be covered with a plastic material. All woven materials are prone to damage and should be regularly checked for serviceability. Slings should be protected from sharp edges and placed evenly about the load, not twisted. Care must be taken to see that the rated capacity (SWL) is not reduced by having a sling angle greater than 90°.

8.12.8 Shackles

Two types of shackle are commonly used in lifting operations. They are the **dee type** shackle and **bow type** shackle, both of which are available with threaded or plain pins.

Only bow type shackles may be used to suspend a load from a hook. The shackle must be positioned with the pin across the hook and the load suspended from the bow. If necessary, spacers should be fitted over the pin to centralise the shackle on the hook.

Overloading, out-of-balance loads and misuse can distort shackles; they should be checked regularly for shape and wear.



Dee shackle



Bow shackle

8.12.9 Eyebolts

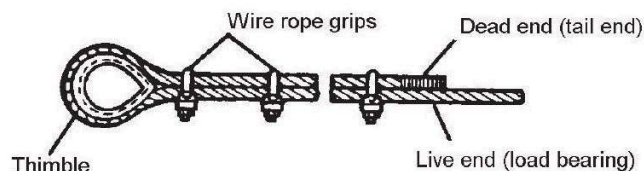
Eyebolts are made to screw into or through a load, and may be plain (dynamo) or have collars, with or without links. The plain eyebolt is good only for vertical loading. Even when a collared eyebolt is used, the rated capacity (SWL) is reduced if the load to which it is attached is initially lifted at an angle.

Collared eyebolts with links may be used providing the angle of load to the axis of eyebolt thread does not exceed 15°. Over 15°, rated capacity (SWL) must be derated in accordance with *Forged steel eyebolts Grade 4* (BS EN ISO 3266) for general lifting purposes. When installed, the collar must be at right angles to the hole, should be in full contact with the surface, and be properly tightened.

The load should always be applied in the plane of the eye, never in the other direction. If necessary, washers or shims should be inserted below the collar to ensure that the eye is correctly aligned when tight. Extreme care must be taken to ensure that metric threaded eyebolts are not inserted in imperial threaded holes. Although these might appear to match, it is an interference fit only, and the mechanical strength may be almost nothing.

8.12.10 Bulldog grips (wire rope grips)

Bulldog grips, if used properly to make an eye with a thimble, provide a simple and effective means of securing the ends of wire ropes instead of splicing or socketing. The final rated capacity (SWL) will be about 75% of that of the wire. It is essential that the correct size and type of grip is used, that the wire is clean and that the correct torque is applied when tightening the grips.



Correct method of fitting wire rope grips

Bulldog grips must be fitted with the 'U' bolt on the dead or tail end of the line (non-load-bearing end). There must be no deviation from this.



The image of the wire rope grips shown above is for illustration purposes only. The number of wire rope grips will change depending on each specific set of requirements.

Wire rope grips meeting the requirements of BS EN 13411 are efficient when correctly installed by a competent person. In all cases, the manufacturer's recommendations should be closely followed, in addition to the advice given in the standard. The correct number of grips must be used and tightened to the correct level of torque.

The wire should be in good condition and all threads and nuts should be suitably greased. The first bulldog grip should be fitted as close as possible to the thimble and, thereafter, at a spacing of no greater than six times the rope diameter.

Nuts must be tightened to the relevant torque:

- when the rope is assembled
- when taken into use
- on the application of the load
- at periodic intervals when in continuous use.

If the above criteria are not followed, then the strength of the eye and thimble will be considerably reduced. If the criteria are neglected for an extended period, the eye and thimble may fail.

Wire rope grips must not be used to make lifting slings, long splices, or to join two wire ropes, other than with thimbles or eyes. For intermediate diameters of rope, the next-largest wire rope grip should be used, except as follows.

- The 5 mm grip should only be used on a nominal 5 mm diameter rope.
- For 11 mm diameter rope, use four 10 mm rope grips tightened to a torque of 14 Nm.

Maximum nominal rope diameter (mm)	Required number of wire rope grips	Tightening torque required (Nm)
5	3	2.0
6.5	3	3.5
8	4	6.0
10	4	9.0
12	4	20.0
14	4	33.0
16	4	49.0
19	4	68.0
22	5	107.0
26	5	147.0
30	6	212.0
34	6	296.0
40	6	363.0

8.12.11 Maintenance, checks, inspections and thorough examinations of lifting accessories



For further information refer to Appendices A and B.



For further guidance on the selection, use, maintenance, inspection and thorough examination of lifting accessories, and planning and carrying out lifting operations with excavators, refer to the Lifting Equipment Engineers Association Code of Practice for the safe use of lifting equipment.

Appendix A – Maintenance, checks and inspections

Maintenance, checks and inspections of lifting equipment are essential if the equipment is not to deteriorate over time, break down and ultimately fail. All maintenance should be carried out by a competent person who is familiar with the equipment and has the knowledge and experience to detect and repair existing or potential faults.

All work should be in accordance with the manufacturer's instructions and be carried out at the recommended intervals.

All lifting equipment should have a pre-use check carried out daily or at the start of each shift and any defects reported to a supervisor. This is normally undertaken by the operator who should be trained to carry out the checks.

In addition to pre-use checks, inspections should be carried out at appropriate intervals, either as part of the maintenance process or at more frequent intervals (such as the weekly inspection of mobile cranes).

The frequency of inspections will depend on assessment of the risk of failure of the lifting equipment and should be guided by the manufacturer's instructions.



Equipment (such as self-erecting tower cranes) often needs to be dismantled for routine inspection and maintenance to be carried out safely and effectively



It is particularly important to ensure at frequent intervals that the load control device or rated capacity limiter on lifting equipment is functioning correctly.

The results of all checks and inspections should be recorded and there should be a system in place to ensure that all reported defects are rectified and recorded.



Each item of lifting equipment should have a history file containing all records of maintenance, inspection and thorough examination. The records can be used to identify maintenance trends and protect against prosecution by demonstrating adequate maintenance in the event of an accident.



For further guidance on maintenance and inspection refer to BS 7121 Part 2-1.

Appendix B – Thorough examination

Types of thorough examination

There are four situations where thorough examination is required by LOLER.

1. Before being put into use for the first time.
2. After installation on a new site, installation in a new location on the same site, or where significant changes have been made to the lifting equipment (such as an extension of a tower crane jib).
3. Periodically while in service.
4. After exceptional circumstances have occurred (for example, extreme weather or impact).

Frequency

LOLER place a duty on employers to ensure that all lifting equipment and lifting accessories are subjected to a thorough examination by a competent (appointed) person at intervals **not exceeding** six months for lifting equipment used for lifting persons and all lifting accessories, and 12 months for other lifting equipment.

The competent (appointed) person carrying out the thorough examination may, however, decide, for a variety of reasons, that the examination should be carried out more frequently (for example, every three months).

A thorough examination must also be carried out after any other event (such as damage or overloading) likely to have affected the safety of any lifting equipment or accessories or where the equipment has been out of use for a long period.

Furthermore, where the safety of lifting equipment depends upon the way it has been installed, as for the many items of temporary lifting equipment used on construction sites (such as hoists and tower cranes), it must be thoroughly examined after assembly and before being put into service at a new site, or a new location on the same site, or after alteration on the same site.

Competent (appointed) person

The competent (appointed) person carrying out thorough examinations must have sufficient practical and theoretical knowledge and experience of the lifting equipment to enable defects and weaknesses to be detected, and their importance in relation to the safety of the equipment to be assessed.

It is essential that the competent (appointed) person is sufficiently independent and impartial to allow objective decisions to be made. This does not mean that competent (appointed) persons must necessarily be employed from an external company. If employers and others within their own organisations have the necessary competence then they can use it.

However, if they do, they must ensure that their in-house examiners have the genuine authority and independence to ensure that examinations are properly carried out and that the necessary recommendations arising from them are made objectively.

This means that thorough examinations of lifting equipment should not be carried out by the person carrying out maintenance or installation of that lifting equipment.

In practice, lifting equipment where the consequence of failure is greatest (such as tower cranes, mobile cranes, MEWPs and hoists) are generally thoroughly examined by a third party.

Reporting

Once a thorough examination has been completed the competent (appointed) person must complete the following.

- Immediately inform both the user and the owner of the lifting equipment of any defect or risk to property or the environment, which in their opinion is, or could become, a danger to persons.
- As soon as is practicable make a report of the thorough examination in writing to both the user and the owner of the lifting equipment.
- Where there is, in their opinion, a defect in the lifting equipment involving an existing or imminent risk of serious personal injury, send a copy of the report as soon as is practicable to the relevant enforcing authority (generally the HSE for construction activities).



Reports of thorough examinations must contain the information contained in Schedule 1 of LOLER, which is reproduced in Appendix C.

(Appendix B continues overleaf)

LIFTING EQUIPMENT – APPENDIX B

Retention of reports

LOLER require that reports of thorough examination are kept for minimum specified periods, depending on the type of thorough examination, as shown in the table below.

Type of equipment	Type of thorough examination	Retention period
Lifting equipment	Before first use	Until the user ceases to use the equipment
Lifting accessories	Before first use	Two years after the report is made
Lifting equipment	After installation at a new site	Until the equipment is removed from site
Lifting equipment and lifting accessories	Periodic	Until the next periodic thorough examination report is made or the expiration of two years, whichever is later
Lifting equipment and lifting accessories	Intermediate inspections	Until the next report is made



Each item of lifting equipment should have a history file containing all records of maintenance, inspection and thorough examination, so that maintenance trends can be established and adequate maintenance demonstrated in the event of an accident.

Testing

LOLER specify that testing, including overload testing, may be considered as part of a thorough examination and should be carried out at the discretion of the competent (appointed) person.

The design of some lifting equipment and the use of high tensile materials mean that damage may be caused by conventional overload tests. Consequently, it is important that the competent (appointed) person carrying out the thorough examination or testing takes account of the manufacturer's instructions and, if necessary, consults the manufacturer before carrying out any test.



Overload testing is not mandatory and is only carried out when required by the competent (appointed) person.



For further guidance on thorough examination refer to BS 7121 Part 2-1 and the HSE document *Thorough examination of lifting equipment: a simple guide for employers (INDG422)*.

Appendix C – Information to be contained in a report of thorough examination (Schedule 1 LOLER)

1. The name and address of the employer for whom the thorough examination was made.
2. The address of the premises at which the thorough examination was made.
3. Particulars sufficient to identify the lifting equipment including, where known, its date of manufacture.
4. The date of the last thorough examination.
5. The rated capacity (SWL) of the lifting equipment or (where its rated capacity (SWL) depends on the configuration of the lifting equipment) its rated capacity (SWL) for the last configuration in which it was thoroughly examined.
6. In relation to the first thorough examination of lifting equipment after installation or after assembly at a new site or in a new location:
 - that it is such a thorough examination
 - (if such be the case) that it has been installed correctly and would be safe to operate.
7. For a thorough examination of lifting equipment other than a thorough examination relating to the circumstances in 6 above:
 - whether it is a thorough examination:
 - within an interval of six months under regulation 9(3)(a)(i)
 - within an interval of 12 months under regulation 9(3)(a)(ii)
 - in accordance with an examination scheme under regulation 9(3)(a)(iii)
 - after the occurrence of exceptional circumstances under regulation 9(3)(a)(iv)
 - (if such be the case) that the lifting equipment would be safe to operate.
8. In relation to every thorough examination of lifting equipment:
 - identification of any part found to have a defect, which is or could become a danger to persons, and a description of the defect
 - particulars of any repair, renewal or alteration required to remedy a defect found to be a danger to persons
 - in the case of a defect which is not yet but could become a danger to persons:
 - the time by which it could become such a danger
 - particulars of any repair, renewal or alteration required to remedy it
 - the latest date by which the next thorough examination must be carried out
 - where the thorough examination included testing, particulars of any test
 - the date of the thorough examination.
9. The name, address and qualifications of the person making the report; whether they are self-employed or, if employed, the name and address of their employer.
10. The name and address of a person signing or authenticating the report on behalf of its author.
11. The date of the report.

Appendix D – MEWP handover and inspection report guidance

These guidance notes are provided to assist the people appointed by the site manager to complete reports.

The term *MEWP* covers pedestrian-controlled, self-propelled and power-operated mobile elevating work and access platforms. The types of plant typically covered by the MEWP handover report are as follows.

- Scissor lifts.
- Telescopic boom or jibs.
- Articulating and telescopic booms.
- Towable units.
- Vehicle mounted.
- Self-propelled, or pedestrian controlled.

In the case of plant provided by a plant hire company, at the site manager's discretion, a competent (appointed) person appointed by the plant hirer may conduct the inspection and complete the report. In all other cases the inspection must be carried out by the contractor's competent (appointed) person.

In the case of plant owned or hired by a contractor, a competent (appointed) person employed by the contractor may conduct the inspection and complete the report.

All completed MEWP handover reports shall be retained for inspection on site.

The report is produced in a checklist format for ease of use and is intended to provide a means by which the condition of all plant delivered to site can be checked by the competent (appointed) person. All questions shall record an answer, with a suitable comment or action included when necessary.

The competent (appointed) person is required to confirm the validity of relevant statutory documentation (such as test certificates) and thorough examination and inspection reports.

An aide-memoire is provided below to assist the competent (appointed) person on statutory requirements that apply to certain items of plant.

MEWP	Regime	Comments	Documentation
All types of mobile elevating work platform (MEWP).	Thorough examination: ■ current six monthly (LOLER) ■ after 1,000 operational hours or as stated in the manufacturer's guidance, if different ■ after major repair or modification ■ following an incident. <i>Note: conducted by a competent engineering surveyor/fitter, with reports kept for two years.</i>	■ Fitted with suitable devices to prevent persons being crushed, trapped, struck or from falling when riding or working from carrier. ■ Provision of suitable devices to prevent risk of a carrier failing. ■ Safe means of escape for persons trapped in a carrier.	Evidence of six-monthly thorough examination, including certification of a rated capacity (SWL).
	Weekly maintenance inspection conducted by a competent operator or fitter.		Evidence of weekly inspection.
	Daily (pre-use) visual inspection conducted by a competent operator.		Evidence of daily inspection.



Following completion of the report at receipt of the MEWP on site, formal inspection must be recorded weekly in normal operating conditions or, under higher-risk conditions, daily.

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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

The erection of steel structures can be difficult and dangerous. The principal hazard and cause of injury is falling from height while working from, or gaining access to, working positions. Other hazards include structural instability during erection, lifting operations and the movement of mobile plant in work areas.

Reading this chapter will not give you the competence to erect structures. It will inform you of the common hazards and how, through practical measures, the risks can be effectively controlled, enabling the work to be carried out safely.

9.1 Introduction

The building and construction industry as a whole has an unacceptably high accident rate, and workers handling and erecting steelwork suffer one of the highest fatality and serious injury rates within the industry. The majority of fatal accidents during the erection of steel structures are as a result of falls from height. In addition, many serious accidents occur due to workers being struck by falling materials.

The British Constructional Steelwork Association (BCSA), the national organisation for the steel construction industry, has reported that the increased use of mobile elevating work platforms (MEWPs) has reduced the need to put operatives onto the steelwork, resulting in a significant reduction of falls and fall-arrest accidents reported to the Health and Safety Executive (HSE). The BCSA provides information, directories, resources and a free encyclopedia for UK steel construction information, containing freely-available information for all sectors of steel construction.



For further information on BCSA visit its website.



Using a mobile elevating work platform (MEWP) in the erecting of framework

9.2 Important points

- Most projects involving erecting steelwork will also involve working at height and exposure to associated potential hazards.
- Designers should consider fitting steelwork with proprietary edge protection before it is lifted into place as a collective fall prevention measure.
- Designers should ensure that the erection of steelwork can be carried out safely (for example, designing-in lifting eye attachment points or specifying ground-level fabrication).
- A detailed schedule should be developed and implemented to ensure the safe erection and stability of the structure to completion.
- All steel erection work must be carried out in accordance with a safe system of work including risk assessments and method statements.
- The erection of steelwork should be carried out in such a way that those completing the work are on a stable working platform at all times, for example, a scaffold or a MEWP. Operatives having to go onto the steelwork should be a last resort.
- Ground conditions must be assessed to confirm suitability to take the weight of MEWPs, cranes and their loads.
- The erection of steelwork will usually involve extended crane operations, the use of MEWPs in and around the new structure, plant movements, steel laydown areas and possibly a vehicle park, all of which must be planned for when the site is first set up.
- All lifting operations, including the use of cranes and MEWPs, must be properly planned, carried out by, and under the control of, competent (appointed) persons (lift planner).
- The presence of overhead power cables, other overhead hazards and the possible need to manually align steelwork components for connection at height should be considered at the planning stage and controlled throughout the project.

Duty holders must:

- avoid work at height where they can
- use work equipment or other measures to prevent falls where they cannot avoid working at height
- where they cannot eliminate the risk of a fall, use work equipment or other measures to minimise the distance and consequences of a fall, should one occur.



Falls from height are the biggest cause of fatal accidents during the erection of steel structures.



For additional information and resources regarding work at height, visit the HSE website.

9.3 Design and planning

Important items that should receive attention during design and planning include the following.

9.3.1 Structural stability

The structure must be stable at all times, from when the first piece of steelwork is put into position until it is completed.

Temporary supports (such as bracing, guys or stays) must be used during the erection of any structure that may be unstable or liable to collapse before it is completed. Additionally, where any work is carried out that is likely to adversely affect the foundations or stability of any existing building or structure (or one under construction), all practicable precautions (such as shoring) must be taken. This should be noted in a temporary works register, and a design brief should be developed.

9.3.2 Temporary structures

Any temporary structure must be of good construction, adequate strength and stability, made of sound materials, free from obvious defects and be properly maintained.



For further information refer to Chapter C04 Temporary works.

9.3.3 Safe means of assembly or making connections

Assembly of steelwork components or making connections should be planned so that erectors can do as much of the work as possible at ground level.

Where erectors have to work at a height, provision must be made for safe means of access to the connecting points and any other working areas. Ideally, work will be carried out from MEWPs operating on a suitable surface. However, where this is not possible, designers should consider the following points.

- Is there adequate working space and a suitable work platform for a crane?
- Connections between steelwork components that are simple to make off site can be assembled or fabricated at ground level to reduce the need to work at height.
- The provision of fixed work platforms and ladders.
- The provision of designated anchorage points for safety nets, fall prevention and arrest devices.

9.3.4 Steelwork components

The size, weight and shape of individual steelwork components will influence safe handling and erecting. Designers will need to consider the following.

- Optimising the length of structural members in an attempt to reduce the number of connections at height.
- Identifying the positions where components should be lifted (marked with pen or paint).
- Identifying and indicating centres of gravity where these are not readily evident (marked with pen or paint).
- Marking components as an aid to identification (also to prevent costly mistakes).
- Calculating the weights of components to assist in ensuring safe crane capacities and the location of cranes.

9.3.5 Safe systems of work and method statements for erecting steelwork

Method statements, as part of the safe system of work, should be written for all steel erection projects. The amount of detail required in the method statement will depend on how complex the work is. The method statement should provide clear evidence that attention has been given to design and planning, as well as to ensuring that the project is completed without risks to health and safety.

A typical safe system of work should include the following information.

- Details of how the project will be managed and how health and safety risks will be eliminated, avoided or reduced.
- Details about the site, including any hazardous features (such as power lines or basements) and what effect these will have on the project.
- The required level of competence of the people undertaking the work.
- The safety precautions to prevent falls from height (for example, arranging for as much assembling as possible to be done at ground level, minimising the number of connections to be made at a height).
- How people will be protected from falling objects (for example, use of screens, fans and debris nets, installation of barriers, exclusion zones and warning notices at ground level).
- Details of plant requirements (such as cranes, MEWPs and other lifting equipment) and the competencies required to operate them.
- Arrangements for the safe receipt, off-loading, storage and handling of steelwork components on site.

FRAME ERECTION

- Details of where and how steelwork will be assembled prior to erection.
- The sequential method of erecting the structure and how stability will be ensured at all times.
- How work such as slinging, lifting, unslinging and the initial and final connecting of steelwork components will be carried out safely.
- The means of providing safe access and a safe place of work by methods such as mobile towers, temporary platforms and walkways.
- Details of the means of communication during lifting operations.
- Any requirement for safety nets, safety harnesses and fall-arrest devices (provisions for design features should be specified, for example, attachment points for ladders, safety nets and fall-arrest devices).
- A contingency plan for dealing with any problems that may arise, including emergencies and rescues.



For further information refer to Chapter A05 Risk assessments, method statements and permits to work.

9.4 Site considerations

Potentially hazardous site conditions should be identified. Some typical examples include the following items.

9.4.1 Overhead power lines

If there are any overhead power lines near the proposed erection site, the local electricity supply company must be consulted. The power lines should be made dead, temporarily rerouted or other suitable precautions (such as overhead cable goalposts) taken to prevent any close approach to, or contact with, live overhead lines.



For further information refer to Chapter D07 Underground and overhead services.



For further information on working near overhead power lines and for additional resources visit the HSE website.

9.4.2 Other contractors and their work

Certain work or processes on, or adjacent to, the site may have the potential to adversely affect the health and safety of workers on site. For example, noxious gases, vapours or dusts may be given off from chimneys, stacks, tank vents and ventilation ducts. These may not cause a problem at ground level but may affect steelwork erectors working at height.

9.4.3 Access and egress

The contractor, or principal contractor on a project involving more than one contractor, should check that all of the vehicles associated with the erection or dismantling of steelwork can access and exit the site safely. It is envisaged that the vehicles will mainly be cranes, MEWPs and delivery vehicles (delivering steel and plant).

A safe location for delivery vehicles to park must be identified and communicated to the steelwork contractor if their vehicles need to wait to be admitted to the site to be unloaded. This area must not be immediately outside the site or on any access road, if being there would create an unacceptable obstruction or other hazards for passing traffic or pedestrians. On larger sites, space should be allocated as a dedicated lorry or trailer park.

On sites where there is no space to park large vehicles, appropriate arrangements (such as a road or lane closure) must be made for delivery vehicles (and possibly mobile cranes) to be parked on the public highway adjacent to the site boundary for the off-loading of steel.

Where there is a loading or unloading bay for only a single vehicle, a strict schedule of delivery times must be written, communicated to the steel delivery contractors and adhered to. In this instance, it is highly advisable to identify a vehicle holding area, to which vehicles that miss their slot can be sent pending the allocation of another unloading slot.

Every effort should be made to avoid the need for vehicles to reverse, particularly out of the site gate and back on to a public road. Where this is not possible, suitable precautions must be taken (for example, the use of one or more vehicle marshaller or an alternative traffic-control system). Separate routes must be provided for pedestrians and vehicles and, where possible, separate entrances.



For more detail on delivery and transport management refer to Chapter C01 Site organisation.

9.4.4 Housekeeping

Construction sites must, so far as is reasonably practicable, be kept in good order and a reasonable state of cleanliness. Platforms, gangways, floors and other places must not be obstructed by waste or loose materials. Projecting nails or similar sharp objects in timber or other materials must be removed or knocked down to prevent injury. Materials must be stacked safely.



For posters that can be used for guidance on preventing slip and trip hazards, visit the HSE website.

9.4.5 Existing site features that require consideration

- Suitability of access and egress routes for vehicles and lifting equipment.
- Ground contamination from previous use of the land.
- Ground-bearing capacity for delivery vehicles, lifting equipment and storage of steelwork.
- Checks for underground services.
- Amount of space required for the safe handling and storage of steelwork.
- Adjacent land uses (for example, railway lines).
- Restricted (crane) oversailing rights.
- Restricted area(s) for vehicle movements.
- Any buildings close to the site that may affect the erection process.
- Any rights of access that may bring members of the public close to the erection site.

9.4.6 Lighting

The following areas must be adequately and suitably lit.

- Every place of work.
- Access to and exits from working places.
- Where lifting operations are in progress.
- All dangerous openings.

9.4.7 Protection from falling material

At any place where people work, steps must be taken to prevent them from being struck by any falling material.

Scaffold components, tools and other objects should be properly lowered to the ground and not thrown or tipped from a height which can cause injury.

9.4.8 Lifting and slinging

All lifting operations should be properly planned by a competent person (lift planner). The level of supervision should reflect the complexity of the operation. Appropriate precautions must be taken to ensure the stability of lifting appliances when used on soft, uneven or sloping ground. These could include measures such as ground levelling, use of mats or hard standing.

The load must not exceed the rated capacity (safe working load (SWL)) of the lifting equipment (allowance should be made for the weight of lifting slings, beams, and so on). Any load that approaches the rated capacity of a crane should be carefully raised a short distance and the operation stopped to check safety and stability before continuing.

If more than one lifting appliance is used to raise or lower a load, levels of supervision should be increased. All lifting appliances should be operated well within their rated capacities to ensure stability. Suitable packing should be used to prevent the edges of the load from damaging the slings.

Loads should be lifted and lowered slowly to avoid any shock loading. The use of tag lines, fixed near the ends of the load, will help to prevent the load spinning when being lifted or lowered. Loads should be lowered onto timber bearers or battens to protect slings and enable them to be removed easily.

For further information refer to:



- Chapter C07 Lifting operations
- Chapter C08 Lifting equipment
- the Lifting Operations and Lifting Equipment Regulations.



For further information and for additional resources, refer to the HSE website and the *Code of practice for safe use of cranes – Part 1: General (BS 7121-1)*.

9.4.9 Off-loading, stacking and storage of steelwork

A safe means of access and a safe workplace must be provided when off-loading components from delivery vehicles.

Two examples of recently developed safety systems are outlined below.

1. Tensioned steel wire running the length of the trailer, at above the head-height of anyone standing on the trailer. Each slinger wears a safety harness and a fall arrest lanyard, the free end of which is clipped to the tension steel wire. In the event of a trip or stumble, the slinger is prevented from falling from the trailer.
2. 'U' shaped inflatable airbags or beanbags that fit around the back and sides of the trailer to act as a soft-landing system, in case anyone falls off the trailer.

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Many accidents have occurred during the off-loading of vehicles when releasing the load-securing mechanism. This is because the load was either not stable when loaded or because it moved and became unstable during the journey.

The stability of the load on the vehicle or in the stack must be ensured at all times. Suitable timber wedges or packing pieces can be used as an aid to stability. Precautions should be taken to prevent slingers being struck by the load (for example, taking up a safe position off the lorry before the load is lifted).

The stack must be constructed so that components can be removed without the risk of someone being trapped or struck. There may be an advantage in using a suitable transportable storage rack (stillage) for smaller components. Anyone not directly involved in the off-loading activity should not be allowed into the area.



Pedestrians and vehicles should be kept apart where practicable.



The HSE leaflet *Safety in the storage and handling of steel and other metal stock (HSG246)* offers practical advice to everyone involved in the delivery and unloading of steel stock.

9.4.10 Traffic routes and vehicles

Prior to, and during, steel erection, deliveries of steel on articulated transport will be a feature of many jobs. In the context of this chapter, Construction (Design and Management) Regulations 2015 place a legal duty on the person in charge of the site to ensure the following.

- That pedestrians and vehicles can move safely and without risks to health, so far as is reasonably practicable.
- There are sufficient traffic routes, suitable for the vehicles using them.
- There is sufficient segregation between pedestrians and moving vehicles, but where this is not reasonably practicable:
 - other means of protection are provided, such as traffic marshals and exclusion zones
 - a means of warning pedestrians of the approach of vehicles, such as vehicle alarms, where the pedestrians would otherwise be at risk, is provided.
- That (vehicle) loading bays have one exit for the exclusive use of pedestrians.
- Appropriate signs are erected in the interests of health and safety.
- Steps are taken to prevent the unintentional movement of any vehicle.
- Each vehicle must be operated in a safe manner with its load arranged securely.
- Every vehicle is fitted with a means of warning people who may be at risk when the vehicle is moving.

9.5 Safe working

The safe system of work must ensure the safety of all operatives:

- whilst getting up to the place of work and down again
- during the period of time at high level before an operative is able to clip on to a means of fall arrest or fall prevention systems.

If access has to be made inside a structural steel box section or any configuration of steelwork where ventilation is poor, it should be treated as a confined space and appropriate precautions taken. The atmosphere should be tested by a competent person, before entry is made, to ensure that there is sufficient oxygen present. Continuous monitoring for oxygen deficiency will also be required.

Tests for flammable or toxic gases or vapours and oxygen enrichment should be carried out as appropriate, depending on the proposed work activity (for example, oxy-fuel cutting or burning may lead to a build up of toxic gases or toxic metal fumes).



For further information refer to Chapter D08 Confined spaces.

All welding fume can cause lung cancer, so you must put controls in place. It can also cause asthma and many other health conditions. The HSE has issued a safety alert about the health risks of welding fume.



For further information visit the HSE website *Welding fume: protect your workers*.

9.5.1 Stability of structures

Steel erection, by its definition, involves the assembly of structures which need to be stable at all times. In particular:

- all practicable steps must be taken to ensure that no person is put at risk by the collapse of any structure that may become unstable or weakened
- structures must not be loaded to an extent that they become unsafe

- any temporary means of supporting a permanent structure must be:
 - designed and maintained to withstand any stresses and strains that are put on it
 - only used for the purpose for which it was designed and installed
 - not overloaded so as to render it unsafe
- there may be a requirement for hold points and the use of permits to work.

9.5.2 Safe means of access and safe place of work

To comply with the Work at Height Regulations, consideration must be given to reducing the need to work at height, for example by:

- doing as much of the connecting work as possible at ground level, or from erected floor decks as the work progresses
- using a releasing device so that lifting gear can be released remotely
- ensuring that, wherever possible, inspection and testing is carried out at low level.

When people must work at height, consideration should be given to the following requirements.

- Installing permanent or temporary handrails, stairways, ladders, walkways and floor decking, so that these can be used by erectors as the work progresses.
- Providing hard standings or floor slabs so that mobile access platforms can be used.
- Providing temporary access and working platforms, including scaffolds, lightweight staging and purpose-built platforms with safe means of access.
- Working from MEWPs.

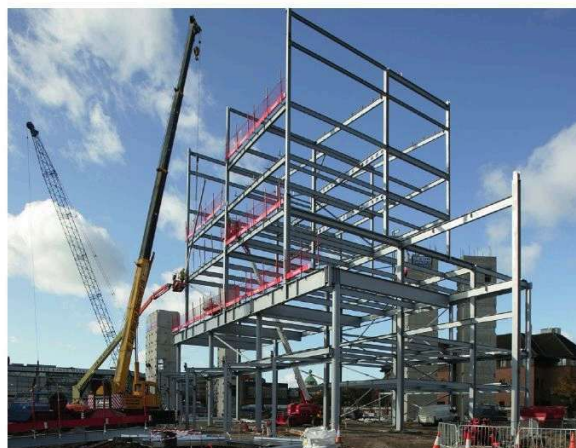
Whilst the increased use of MEWPs has brought about safer working at height, it has also created the need to ensure that MEWP operators are adequately trained and competent, and that ground conditions on the site are properly surveyed and prepared to enable the safe use of such heavy plant without the risk of sinking or overturning.

It may be appropriate in some circumstances to use other means of access to height, such as roped access techniques, which must be carried out by trained and competent persons.

Alternatively, access may be achieved by using a personnel carrying platform suspended from a crane.

There may be occasions when it is necessary to use fall arrest rather than work-restraint measures, with safety nets or other soft-landing systems being the preferred method of fall arrest.

The BCSA has stated that the increased use of safety nets has had a significant beneficial effect on the steel construction industry's accident rate.



Steel erection in progress using a mobile crane, two MEWPs and edge protection being attached to steelwork at ground level before being lifted into position



For further information refer to Chapter D02 Working at height, and Chapter D05 Fall arrest and suspension equipment.

9.5.3 Edge protection

Traditionally, scaffolding was the main form of edge protection and could only be installed, modified or removed by trained scaffolders, after the steelwork was in place. However, the development of various types of system edge protection has provided opportunities for pre-installing edge protection before work at height commences.

The use of such systems can also eliminate the need for successive trades to install their own edge protection and then remove it when leaving site, only for it to be replaced by the next trade's edge protection. The elimination of such duplication has obvious safety benefits.

9.5.4 Beam straddling

There may be occasions where the work cannot be done from a MEWP or other platform and erectors may have to work from the steel frame. This is known as beam straddling. This form of access should only be considered as a **last resort** and only where the beam is of I-beam section.

A full body harness with a twin-tailed lanyard system may be used. However, users must ensure that this system is used correctly and that the second tail is not attached to the user's harness, as this may interfere with the operation of the energy absorber.

Consideration must be made as to suitable anchor points to which the lanyard can be attached. The erector can sit astride the flange with the sole and heel of each foot resting on the bottom flange and with both hands able to grasp either side of the top flange.

FRAME ERECTION

Alternatively, proprietary beam-gliding devices are available to improve mobility. The risk assessment **must** consider how anyone carrying out this practice is going to be rescued following an emergency or a fall. The use of a safety harness or lanyard requires a minimum clearance below the high-level place of work to allow the lanyard to function properly in arresting a fall. Expert advice should be sought.



For further information on safe working at height refer to Chapter D01 Work at Height Regulations and Code of practice for selection, use and maintenance of personal fall protection systems and equipment for use in the workplace (BS 8437:2005).

9.5.5 The weather

A regular weather forecast should be obtained by the person in charge of the erection programme. Weather conditions (such as wind speeds) should also be monitored throughout the work. Erection should not take place where weather conditions impose an adverse effect, for example, from ice, frost, snow, strong winds, rain, dew and poor visibility (such as fog, mist or glare).

If erection work is stopped, measures should be taken to ensure that the structure remains stable. After a stoppage due to the weather, stability of the structure should be checked by a trained and competent person before work is allowed to restart.



This section provides an overview of the steel erecting sector and should not be read in isolation. For comprehensive understanding of the site manager's role it should be read in conjunction with all other sections A, B, C, D, E and F of GE700.

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Mobile workforce and driver safety



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



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Overview

The description of a mobile workforce covers a multitude of trades in many different situations from a lone worker called out to an emergency, with little knowledge and control, to a group of workers operating in an unfamiliar or hostile environment where customer or client control is minimal.

This chapter provides a general overview of the legal framework and some practical guidance on the management of health and safety at work for mobile, lone and out-of-hours workers. It includes information about the potential risks they may be exposed to, and practical advice on the management and control of their work.

10.1 Introduction

This chapter aims to provide employers and managers with general practical guidance on processes and controls that can be put in place to provide their workers with a safe place of work. This ranges from simply saying 'tell us where you are' to planned work that identifies mobile locations where permits may be required. Management preparation information, pre-job briefing, competence, self-discipline, control, awareness and training are also identified, all of which will form a package of safety for the mobile workforce.



Employers are responsible for the health, safety and welfare at work of all of their employees. These responsibilities cannot be transferred to any other person, including those people who work alone. It is the employer's duty to assess risks to lone workers and take steps to eliminate or control risks where necessary.

The types of workers covered within this chapter are mobile groups of workers, lone workers and out-of-hours workers.



Types of workers

- Mobile groups of workers.
 - A group of operatives working away from their home base, undertaking such work as domestic and small-scale commercial refurbishment. This can include painters and decorators, scaffolders and roof workers.
 - A group of workers undertaking minor road works (*refer to Chapter F01 Street works and road works*) and rail maintenance works (*refer to Chapter F02 Trackside safety*).
 - Those working in areas accessed by the general public, completing work for utility companies (gas, electric, water and telecommunications).
- Lone workers.
 - This group will include maintenance workers (such as plumbers, electricians, lift engineers, water and gas specialists, domestic aerial installers, aerial and mast maintenance personnel, and many others).
- Out-of-hours workers.
 - Those identified as mobile groups of workers or lone workers may, in addition, be required to work out of normal working hours to ensure continuance of 24-hour operations or emergency response (for example, security guards or utility company workers – gas, water, electricity and telecommunications).

10.2 Important points

10.2.1 Groups of workers

- Assess the risks to the mobile workers for that particular environment and the tasks to be undertaken.
- Ensure ongoing consultation so that all relevant hazards can be identified.
- Employers with five or more employees have to record significant findings of risk assessments.
- Ensure there is a safe system of work in place prior to work commencing.
- Provide the workers with a means to undertake a dynamic risk assessment and update the risk assessment to create an 'at point of work risk assessment'.
- Confirm the workers have a suitable level of training and competency (such as from the Construction Skills Certification Scheme (CSCS), the Prefabricated Access Suppliers and Manufacturers Association (PASMA), and trade qualifications).
- Decide on equipment needed. Are special transport arrangements required?

- Ensure that any vehicle being used is suitable for the number of personnel to be carried.
- Ensure the vehicle is appropriate if required to tow equipment (such as a mini digger, trailer or welfare unit).
- Ensure the vehicle is rated for the safe towing weight limit.
- Ensure the nominated driver is trained and qualified to tow.
- Consider safe access to the roof of the vehicle (such as mechanised roof access), if it is unavoidable that equipment will need to be carried (such as ladders and pipes).
- Avoid climbing on or off flatbed lorries during deliveries, consider use of pallets or pre-slung loads.
- Does the area need segregation from pedestrians or from other traffic during deliveries?
- Is temporary traffic lighting and street signage needed (*refer to Chapter F01 Street works and road works*)?
- Is a fire emergency plan in place, including details of actions to take, assembly point, audible warning device and fire extinguishers?
- Ensure the work area is secure, including storage of materials, suitable waste containers, and so on.
- Consider what welfare arrangements are needed for these works (for example, provision of an integral welfare unit that complies with Health and Safety Executive (HSE) requirements (refreshment/seating area, toilet, shower, clothes storage and drying, and generator), use of client's facilities (with written agreement) or public facilities within suitable proximity).
- Do the works require an emergency rescue plan – are they high risk, confined space, work at height or electrical?
- Has a first-aid kit been provided, suitable for the number of persons and risks present?
- Is a member of the group trained either as an appointed person, emergency first aider or fully qualified first aider, depending on the outcome of the first-aid needs assessment (*refer to Chapter B05 First aid*).
- Have suitable containers been provided to store waste for removal to suitable disposal facilities?

10.2.2 Lone workers



The HSE defines a *lone worker* as someone who works by themselves, without close or direct supervision.

Lone workers include people who:

- work from a fixed base but are alone on the premises (such as security guards)
- work separately from others on the same premises (for example, security staff), or work outside of normal hours
- work away from a fixed base (such as maintenance workers, health care workers and environment inspectors), work at home (home workers) and mobile workers (such as site managers dealing with multiple sites).

The following points are additional to the relevant items listed for groups of workers (*refer to 10.2.1*).

- Does the lone worker require extra risk control measures (such as a personal alarm linked to a 24/7 alarm-receiving centre, or phone)?
- Has training been given in the completion of the dynamic or point-of-work risk assessment?
- Does the workplace present a specific risk to the lone worker (such as vulnerable tenants or working at height)?
- Is there a safe way in and out of the premises for one person?
- Can any necessary temporary access equipment (such as portable ladders or trestles) be safely handled by one person?
- Can all the machinery and goods involved in the workplace be safely handled by one person?
- Are any chemicals or hazardous substances being used that may pose a risk to the worker?
- Does the work involve lifting objects too large for one person?
- Is more than one person needed to operate essential controls for the safe running of equipment or workplace transport?
- Is there a risk of abuse or violence?
- Are young, pregnant or disabled workers particularly at risk if they work alone and is a further specific risk assessment required?
- Are there any other reasons why the individual (for example, on medication) may be more vulnerable than others?
- Are emergency contact arrangements in place in case of an incident (such as regular reporting to supervisor, a GPS systems/tracker, regular phone contact, monitoring of staff security systems and so on)?



For further information on the risks of working alone refer to the HSE leaflet *Protecting lone workers – how to manage the risks of working alone* (INDG73).

Updated guidance also covers how to protect lone workers from the risk of work-related violence; provides more information on how managers should keep in touch with lone workers; and gives new advice on the impact that lone working can have on stress, mental health and wellbeing.

10.2.3 Out-of-hours workers

Either lone workers or groups of workers may need to work out of hours, frequently at short notice. Therefore, the following points also apply, in addition to those above.

- Are arrangements made for emergency situations (such as who to contact and backup information)?
- Has a specific risk assessment been completed?
- Is a safe system of work in place and has the operative been provided with a method statement for the tasks to be undertaken?
- Do they have the correct equipment required, and can it be operated by the operative(s)?

10.3 Legislative requirements

Many pieces of legislation are applicable to these types of workers – some of these are listed below.

Health and Safety at Work etc. Act 1974 – the umbrella Act under which most health and safety regulations have evolved. For example, it details information regarding an employer's duties in the workplace; access and egress; provision and suitability of plant; information, instruction, supervision and training of personnel; an employee's duties and much more (*for further information refer to Chapter A01 Health and safety law*).

Management of Health and Safety at Work Regulations 1999 – provide guidance on risk assessments and controls, monitoring and review (*for further information refer to Chapter A05 Risk assessments, method statements and permits to work*).

Provision and Use of Work Equipment Regulations 1998 – provide testing and maintenance information (such as chainsaw, angle grinder and compressor) (*for further information refer to Chapter C05 Work equipment and hand-held tools*).

Lifting Operations and Lifting Equipment Regulations 1998 – provide information on cranes, telehandlers, lorry-loaded cranes and lifting accessories, including testing and maintenance requirements (*for further information refer to Chapter C07 Lifting operations and Chapter C08 Lifting equipment*).

Construction (Design and Management) Regulations 2015 – provide guidance for welfare arrangements (*for further information refer to Chapter A03 Construction (Design and Management) Regulations 2015*).

Work at Height Regulations 2005 – provide information on scaffolding, use of ladders and alternative means of working safely at height (*for further information refer to Chapter D01 Work at Height Regulations*).

Personal Protective Equipment at Work (Amendment) Regulations 2022 – information on legislative requirements, for example, types and uses of specific equipment (such as eye protection and respiratory protective equipment) in particular environments and training in the fitting of such items (*for further information refer to Chapter B06 Personal protective equipment*).

10.4 Risk assessment and control



A risk assessment is an important step in protecting your workers and your business, as well as complying with the law. It helps you focus on the risks that really matter in your workplace – the ones with the potential to cause real harm.

Employers need to identify the potential hazards faced by workers and assess the risks involved both to these workers and to any people who may be affected by their work. Employers should ensure that measures are in place to avoid or control such risks. The risk assessment must also identify the level of supervision required. High-risk activities may require another person to be present (for example, if visiting a vulnerable tenant or for working on or near to live electrical equipment).

10.4.1 Employer responsibilities

- **Involve** staff or their representatives when undertaking the risk assessment process.
- Take steps to **check control measures** are in place (examples of control measures include exclusion zones, barriers, use of less hazardous substances and personal protective equipment (PPE), as well as training and supervision).
- **Review** risk assessments and method statements regularly. Most workplaces are subject to changes that can affect existing working practices and introduce new hazards (*also refer to 10.4.2 Point-of-work risk assessments*).
- **Allow** lone working **only** where the risk assessment and method statement shows it is safe to do so.
- Where a lone worker is working at another employer's workplace, that employer should **inform the lone worker's employer** of any hazards and the required control measures.



Refer to the HSE publication *Risk assessment – A brief guide to controlling risks in the workplace* (INDG163).

10.4.2 Point-of-work risk assessment

A risk assessment completed by the employer or supervisor should be provided to workers before any works commencing, where practicable.

However, there are many occasions (as in the examples below) when a point-of-work risk assessment will be required.

- To update the previously prepared risk assessment.
- As a result of changes to the working environment.
- As a result of changes to the works to be undertaken.
- For out-of-hours working, when each call out will require a specific risk assessment (for example, emergency maintenance (such as a serious water leak or flooding) or urgent utilities works (such as a gas leak or explosion)).
- For a maintenance worker who is undertaking a number of different calls in a day where generic information is available but needs to be tailored to a specific location or task.

On arrival at site report to the person on site, contact base to confirm arrival (lone worker safety protocol) and prepare a point of work risk assessment.



Employers and supervisors must satisfy themselves that the workforce is competent to carry out a point-of-work risk assessment. Training should be provided that includes practical, on-site mentoring. If a site activity *cannot* be done safely, the workforce must know when *not to start*, and contact their supervisor for further advice.

Some examples of unsafe working areas and/or conditions are shown below.

- An unguarded excavation.
- Traffic not segregated.
- Permits to work and disconnections and isolations where certificates do not match.
- Scaffolding where inspection records are not present or the date has expired.
- Inadequate or incorrect equipment and lack of training on the equipment.
- No fire extinguisher.
- Incorrect tools.
- Inadequate and/or unsuitable PPE.

10.4.2.1 Filling in a point-of-work risk assessment

Part 1. Stop and ensure the area is safe with adequate access and egress. Complete items listed under this section. Seek advice from a supervisor if you answer **no** to any of them.

Part 2. Think about what hazards may be present and carry out a safety assessment of those identified. Determine if the existing controls are adequate. Provide brief details of your rescue plan for safe escape in the event of something going wrong. Make a note on the form of any hazards that are significant but without, or with insufficient, controls.

Part 3. Act. Detail additional hazards and control measures needed to manage the risk. Is the remaining risk acceptable for work to commence? If no, this needs input from your supervisor.

Part 4. Review, and inform your supervisor if the job created any additional hazards or if there are lessons to be learned.

10.4.3 Dynamic risk assessment

A dynamic risk assessment is a mental evaluation of the environment you are going into. For lone workers, this could be someone's home or a remote location. In these situations, the worker needs to be continually aware of any changes in their immediate environment, make judgments and take appropriate actions to stay safe.

This could include the assessment of people's aggressive behaviour or tone of voice, as well as dangerous physical issues (such as animals or changing weather conditions). This does not replace the need for a full risk assessment but is an important skill for lone workers to have.

10.5 Work-related violence

People who deal directly with the public may face aggressive or violent behaviour. They may be sworn at, threatened or even attacked.



Violence at work is defined as any incident in which a person is abused, threatened or assaulted in circumstances relating to their work.

Lone workers are at greater risk of violence whilst at work. There were an estimated 688,000 incidents of violence at work, as noted in the HSE's report *Violence at work, 2019/20*.

10.5.1 People at risk

People who are employed in jobs (such as those below) that involve them dealing with the public can be at a greater risk from violence.

- Giving a service.
- Education.
- Delivery or collection.
- Controlling.
- Representing authority.

MOBILE WORKFORCE AND DRIVER SAFETY

Both the employer and employees have an interest in reducing violence at work. For employers, violence can lead to poor morale and a poor image for the organisation, making it difficult to recruit and retain staff. It can also mean extra cost, with absenteeism, higher insurance premiums and compensation payments. For employees, violence can cause pain, distress and even disability or death.

Physical attacks are obviously dangerous but serious or persistent verbal abuse or threats can also damage employees' health and wellbeing through anxiety or stress.

10.5.2 Managing violence in the workplace

A straightforward, four-stage management process is listed below.

1. Find out if you have a problem.
2. Take action.
3. Delivery or collection.
4. Check what you have done.

 For further information on work-related violence visit the HSE website.

10.6 Welfare facilities for mobile workers

Health and safety legislation requires that welfare arrangements for projects of any size are prearranged prior to works commencing on site. For these circumstances there are two types to be considered.

1. Use of client's facilities – prearranged and confirmed in writing, or local public convenience. It should not be assumed that local commercial premises can be used without their permission.
2. Provision of an integrated hygiene unit, which complies with HSE requirements and includes a refreshment area, toilet, drying area and generator.

If local facilities or a portaloo are considered sufficient then van-mounted equipment could be used for washing and the preparation of food and refreshment, using a battery-powered inverter.

e.g. Mobile work vans fitted out with welfare facilities



 Visit the HSE website for further information about the provision of welfare facilities during construction work.

10.7 Waste storage and disposal

Waste that is created at a site must be disposed of appropriately. Hazardous waste will need to be stored in appropriate containers and removed by licensed contractors using a hazardous waste consignment note.

Non-hazardous waste generated during works undertaken by mobile groups of workers or lone workers can be transported to their base for sorting and transferring to suitable recycling facilities or landfill sites. The following points will need to be considered.

- Is the waste suitable for transporting? Is it hazardous or non-hazardous, and does it require special handling?
- Is the person removing it competent and authorised to do so? (This includes lone workers who are transporting the waste back to the depot for future disposal).
- Has the waste been segregated (for example, plasterboard separated from biodegradable waste)?
- Can the waste be segregated in the vehicle, by bagging or being placed in enclosed containers, or does it require collection from the location?
- Do the mobile workers have suitable containers to package the waste products?
- Can the waste be carried without exceeding the safe working load of the vehicle?



If using a company vehicle to carry waste back to a depot or to a licensed waste treatment or disposal facility, then a waste carrier's licence may be required.



For further information, refer to Chapter E10 Waste and material management.

10.8 Driver safety

The nature of mobile working makes travel a necessity either between sites or operating plant. The HSE has made it clear that health and safety law applies to driving at work. Therefore, employers have a responsibility to ensure they have considered work-related road risks and implemented the necessary controls.

10.8.1 Fatigue

Another factor to consider is the effects of driving when drowsy. The vast majority of people believe that drowsy driving is unacceptable, yet the incidence of drowsy driving suggests there is a disconnect between what people say and what they do. Depending on which piece of research you read, between 20% and 40% of all accidents on the road are attributable to fatigue. Companies should encourage a healthy work-life balance and be alert to signs of driver fatigue.

Research has been undertaken to investigate the costs and understand the signs associated with drowsy driving, and to provide strategies to prevent it and explain how organisations can counter the problem.



For further information, visit the Third Pillar of Health website.



Details of safe driving and work-related road risk can be downloaded from the Prospect website.

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SITE SAFETY PLUS

Construction site health and safety

C: General safety

This section covers some common elements of construction site safety.

It contains important information and guidance on site organisation, safety signs, fire and emergency arrangements, electrical safety, temporary works awareness, the provision and safe use of plant and equipment, managing mobile plant, lifting operations and equipment, frame erection, and mobile workforce and driver safety.

Ensuring these common elements are effectively managed helps sustain safe and tidy sites, reduces accidents and ill health, and maintains productivity.